Introduction

1.1 Purpose of this Document

This Software Requirements Specification (SRS) establishes the definitive technical, operational, and functional baseline for the **Autometa School Enterprise Resource Planning (ERP) Platform** (`baseline_V3`).

The document serves as the single source of truth across the entire Software Development Life Cycle (SDLC):

- **Engineering & Architecture:** Provides backend, frontend, and database engineers with precise requirements, business logic, multi-tenant boundaries, and data processing rules.
- **Quality Assurance & Verification:** Serves as the baseline for constructing test matrices, validation suites, and Behavior-Driven Development (BDD) scenarios.
- **Project & Scope Governance:** Defines clear feature cutoffs, distinguishing minimum viable product (MVP) scope from extended or subsequent rollout phases.
- **Regulatory & Administrative Sign-off:** Provides school leadership, legal auditors, and institutional administrators with validation criteria verifying compliance with statutory education board guidelines and national data protection mandates.

1.2 Scope of this Document

This document covers the end-to-end design, system boundaries, and execution requirements for a cloud-native, multi-tenant school operating system supporting diverse educational boards (CBSE, ICSE, IB, and State Boards).

In-Scope Functional Boundaries

- **Core Academic & Operational Modules:** Admission & enquiry pipelines, 360° student lifecycle tracking, staff/HR records and payroll, multi-board curriculum design, timetable scheduling and substitution engines, and examination/gradebook generation.
- **Offline-First Field Operations:** Progressive Web Application (PWA) architecture utilizing Service Workers and IndexedDB for uninterrupted offline attendance capture, offline grade entry, and automated background reconciliation.
- **Financial & Audit Governance:** Double-entry ledger accounting, flexible fee structure configurations, payment gateway hooks, and immutable financial logging.
- **Multi-Portal Ecosystem:** Dedicated, role-tailored web/PWA portals for Administrators, Teachers, Parents, and Students with scoped data visibility and role-based access control (RBAC).
- **Communication & Notification Dispatch:** Multi-channel alerting pipeline integrating Firebase Cloud Messaging (FCM), in-app notifications, WhatsApp Business API, and SMS fallback with rate and cost caps.
- **Regulatory Compliance:** Architectural alignment with the **Digital Personal Data Protection (DPDP) Act, 2023**, board regulations, and ISO/IEC 27001 data protection principles.

Out-of-Scope / Deferred Boundaries

- Standalone hardware telematics (GPS hardware firmware/manufacturing; system consumes standardized telemetry feeds only).

- Full LMS content creation engines (advanced SCORM authoring deferred; MVP limited to LMS-Lite homework distribution).
- Custom third-party legacy accounting integrations beyond standard CSV/API double-entry export formats in Phase 1.

1.3 Overview

The **Autometa School ERP** is engineered as a secure, distributed multi-tenant software-as-a-service (SaaS) platform built to resolve administrative overhead, eliminate communication gaps, and maintain absolute data integrity for modern primary and secondary institutions.

The remainder of this document is organized into structured sections:

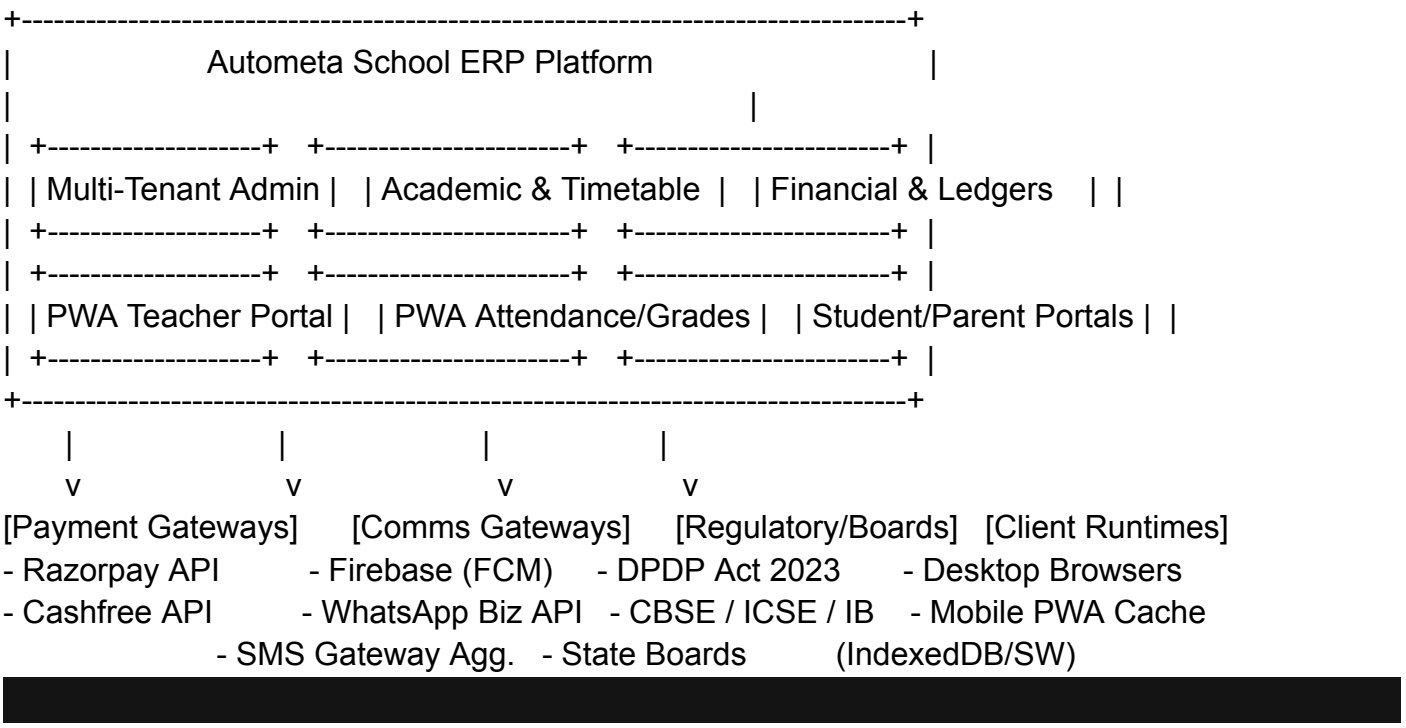
- **Section 2 – Overall Description:** Defines user personas, real-world operating environments, system constraints, deployment dependencies, and critical architectural pre-requisites (schema isolation, prototyping, and offline persistence).
- **Section 3 – System Features and Specific Requirements:** Details granular functional requirements using structured syntax, covering each module from administrative dashboards and automated timetable engines to parent/student portals and ledger engines.
- **Section 4 – External Interface Requirements:** Specifies UI/UX interaction standards, hardware constraints, database schemas, and external API gateways (Razorpay, Cashfree, WhatsApp API, FCM).
- **Section 5 – Non-Functional Requirements & Compliance:** Details system reliability, performance thresholds, offline synchronization conflict resolution, Failure Modes and Effects Analysis (FMEA), ISO 27001 security standards, and strict adherence to DPDP Act privacy principles.

General Description

2.1 Product Perspective

The **Autometa School Enterprise Resource Planning (ERP) System** (baseline_V3) is a standalone, cloud-native Software-as-a-Service (SaaS) platform built to manage academic, administrative, operational, and financial functions across individual campuses and federated multi-campus groups.

The system replaces fragmented single-purpose tools (discrete spreadsheets, legacy desktop databases, and third-party messaging groups) with a single multi-tenant backbone. It operates in an ecosystem bounded by external services and regulatory bodies:



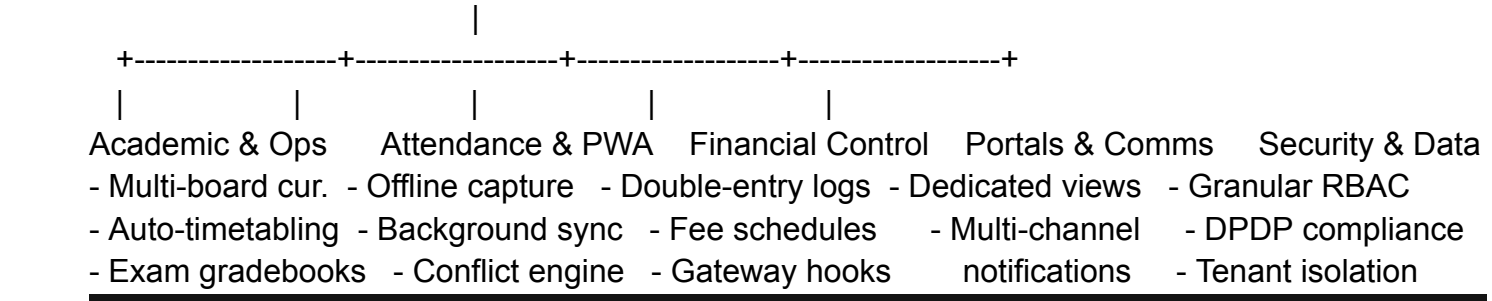
2.1.1 System Interfaces

- **Payment Processing:** Secure hooks into Payment Gateway APIs (Razorpay, Cashfree) supporting webhook-driven transaction reconciliation for online fee collection.
- **Communication Pipelines:** External integration with Firebase Cloud Messaging (FCM) for push notifications, WhatsApp Business API for direct broadcast alerts, and SMS fallback routes governed by budget caps.
- **Telematics Data Ingestion:** Standardized webhook and REST ingestion endpoints for third-party vehicle GPS hardware tracking, consumed by the Transport Management module.

2.2 Product Functions

The platform automates institutional workflows across five primary functional domains:

Autometa Core Capabilities



- **Academic & Timetable Automation:** Automated master grid generation, soft/hard constraint scheduling, teacher workload balancing, and dynamic same-day teacher substitution calculation.
- **Offline-First Field Operations:** High-speed attendance capture and exam marks recording via Progressive Web Application (PWA) with background sync upon reconnection.
- **Financial Ledger Management:** Double-entry accounting system tracking dynamic fee structures, concession approvals, automated invoice dispatch, fine calculations, and audit-ready fiscal reconciliation.
- **Stakeholder Portal Ecosystem:** Segmented operational portals (Administrator, Teacher, Student, Parent) delivering real-time academic, attendance, and transport tracking.
- **Administrative Governance & Compliance:** Native data governance adhering to the Digital Personal Data Protection (DPDP) Act 2023, automated student document lifecycle generation (Transfer Certificates, Bonafide), and multi-campus system administration.

2.3 User Classes and Characteristics

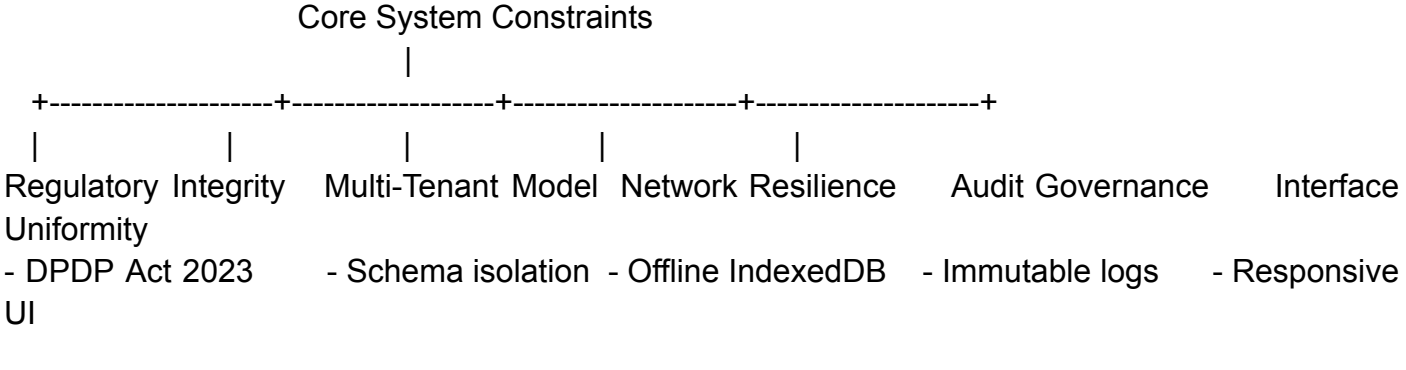
User Class	Technical Expertise	Operational Context	Primary Responsibilities & Access Scope
System Super Admin	Advanced	Desktop web	Configures multi-tenant parameters, provisions campus nodes, assigns role permissions, audits system logs.
Campus Principal / Administrator	Intermediate	Desktop web	Supervises school-wide operations, approves fee concessions, reviews audit dashboards, publishes official notices.
Academic Coordinator	Intermediate	Desktop web	Configures board curriculum, manages master timetables, handles teacher allocations, initiates substitution workflows.

School Accountant / Fee Clerk	Intermediate	Desktop web	Sets fee structures, manages ledger balances, processes offline/cash receipts, reconciles payment gateway transactions.
Teacher / Academic Staff	Basic to Intermediate	Mobile PWA & Desktop	Records attendance offline/online, logs exam grades, views assigned timetables and substitution alerts.
Parent / Guardian	Basic	Mobile PWA / Web	Tracks ward attendance, pays school fees, monitors academic grade reports, views transport telemetry, manages consent.
Student	Basic to Intermediate	Mobile PWA / Web	Inspects class schedules, reviews homework assignments, accesses report cards, tracks issued library books.

2.4 Operating Environment

- **Server-Side Infrastructure:** Containerized Linux runtime (Ubuntu 22.04 LTS / Alpine-based container images), orchestrated via Kubernetes or Docker Swarm, supporting auto-scaling web application nodes.
- **Database Management System:** PostgreSQL 15+ configured with logical schema-based isolation per tenant, leveraging read replicas for asynchronous report rendering.
- **Client-Side Desktop Targets:** Modern Evergreen web browsers (Google Chrome 110+, Mozilla Firefox 115+, Microsoft Edge 110+, Apple Safari 16+).
- **Client-Side Mobile/PWA Targets:** Android 9.0+ (Chrome / PWA Shell), iOS 16.4+ (Safari PWA with push notification support and Service Worker capabilities).
- **Local Storage & Hardware Constraints:** Minimum 512 MB available local device storage on client hardware for PWA caching, local IndexedDB persistence, and asset synchronization.

2.5 Design and Implementation Constraints



- Parental consent - PostgreSQL rows - Bi-directional sync - Double-entry rules - PWA compliance

- **Regulatory & Statutory Integrity:** Strict adherence to India's DPDP Act 2023 regarding the collection, storage, and processing of minor data, requiring auditable parental consent and manual data retention/erasure policies.
- **Multi-Tenant Architecture:** Strict database isolation using PostgreSQL schema-based separation to prevent cross-tenant data leaks across distinct educational institutions.
- **Network & Offline Resilience:** Field modules (Attendance, Grading) must function without active internet connectivity; client runtimes must manage transaction queues via IndexedDB and handle sync conflicts using deterministic server-authoritative timestamps.
- **Financial Integrity:** Fee management operations must conform to strict double-entry ledger bookkeeping; ledger entries cannot be overwritten or hard-deleted, requiring offsetting credit/debit entries for adjustments.
- **Interface Uniformity:** All portal interfaces must follow responsive web design standards to maintain usability across varying screen resolutions (360px mobile viewports to 1920px desktop displays).

2.6 User Documentation

The platform ships with operational manuals and onboarding resources:

- **System Administrator Guide:** Detailed technical documentation covering tenant provisioning, schema migrations, backup and recovery setups, and RBAC matrix configurations.
- **Institutional User Manuals:** Role-specific workflows for fee clerks, academic coordinators, and teachers, detailing timetable generation, ledger balancing, and offline PWA operation.
- **Self-Service Knowledge Base:** In-app tooltips, contextual help documentation, and downloadable PDF onboarding sheets for parents and students explaining portal access, online fee submission, and multi-child profile switching.

2.7 Assumptions and Dependencies

- **Browser Feature Availability:** Offline capability assumes client devices run modern browsers fully supporting Service Workers, the Cache API, and IndexedDB v2.0+.
- **Identity Verification Services:** Parent and staff onboarding assumes regular uptime and operational availability of SMS-based and Aadhaar-linked OTP gateway verification services.
- **Payment Gateway Availability:** End-to-end fee clearance workflows depend on operational uptime and timely webhook dispatch from configured payment service providers (Razorpay, Cashfree).
- **Third-Party Telemetry Streams:** Live vehicle transit updates rely on the availability, signal continuity, and API compliance of third-party GPS hardware transponders installed on school transport fleets.

System Features and Specific Requirements

3.1 Functional Requirements Matrix

The following table details all 21 functional requirements for the Autometa School ERP Platform (baseline_V3), specifying their rollout priority, MVP scope, core functional capabilities, and primary stakeholders.

Ref ID	Feature Name	MVP Status	Priority	Core Functional Scope & Technical Capabilities	Primary Stakeholder
FR-01	**Dashboard**	In MVP	High	Role-specific, real-time interactive widgets and KPI visualizations covering daily attendance trends, fee collection summaries, academic milestones, and operational metrics.	Super Admin, Principal, Accountant, Teacher
FR-02	**Admission & Enquiry**	In MVP	High	Lead capture pipeline, prospect stage tracking, registration forms, parent mobile OTP/identity verification workflows, and DPDP-compliant explicit parental consent collection.	Admissions Office, Parents, Principal
FR-03	**Student Management**	In MVP	Critical	360-degree student lifecycle profile management, demographic records, medical history, academic progression history, disciplinary logs, and section enrollment tracking.	Admin, Teachers, Parents

FR-04	**Staff / HR Management**	In MVP	High	Employee lifecycle management, credential storage, department/designation mapping, biometric/PWA staff attendance, leave balance ledgers, and automated monthly payroll calculation.	HR Manager, Accountant, Principal, Staff
FR-05	**Academic Management**	In MVP	Critical	Multi-board academic framework configurations (CBSE, ICSE, IB, State Boards), multi-term academic calendar definition, subject-to-class mapping, and section allocation rules.	Academic Coordinator, Principal, Teachers
FR-06	**Timetable Management**	In MVP	Critical	Manual master schedule entry, hard/soft constraint and period collision validation (no teacher/room overlaps), facility binding, real-time teacher substitution engine, class-wise and teacher-wise timetable generation, and multi-portal schedule syncing.	Academic Coordinator, Teachers, Students

FR-07	**Attendance Management**	In MVP	Critical	Offline-first Progressive Web Application (PWA) attendance capture via IndexedDB, automated background synchronization, period/daily attendance modes, and instant absent parent alerts.	Teachers, Admin, Parents
FR-08	**Homework & Assignment**	Optional / Phase 2	Medium	LMS-Lite repository for homework publication, multimedia attachments, student digital submissions, teacher evaluation workflows, and deadline tracking notifications.	Teachers, Students, Parents
FR-09	**Examination & Results**	In MVP	High	Assessment scheduling, offline-capable marks entry sheets, multi-board grading scale calculators (CCE/marks/GPA), audit-locked gradebooks, and automated batch PDF report card generation.	Exam In-Charge, Teachers, Students, Parents

FR-10	**Fee Management**	In MVP	Critical	Dynamic fee structure definitions (tuition, transport, lab), double-entry transaction ledgers, concession workflows, payment gateway webhooks (Razorpay, Cashfree), and automated digital receipts.	Accountant, Cashier, Admin, Parents
FR-11	**Parent Portal**	In MVP	High	Scoped multi-child dashboard for parents to monitor daily attendance, pay tuition fees, download fee receipts, inspect exam progress reports, view transport telemetry, and exercise DPDP privacy rights.	Parents
FR-12	**Student Portal**	In MVP	Medium	Dedicated read-mostly view for class timetables, daily homework assignments, published exam results, institutional announcements, and issued library book tracking.	Students
FR-13	**Teacher Portal**	In MVP	Critical	Responsive, offline-first PWA workspace enabling class roll-call marking, grade entry, personal timetable and substitution views, and direct parent	Teachers

				messaging within administrative bounds.	
FR-14	**Library Management**	Extended Scope	Low	ISBN cataloging, physical barcode/QR scanning, circulation desk operations (issue, renewal, return), overdue fine calculation engines, and lost item depreciation tracking.	Librarian, Students, Teachers
FR-15	**Transport Management**	Extended Scope	Medium	Fleet inventory, vehicle maintenance logs, bus route and pickup stop optimization, student route mapping, driver licensing records, and external GPS telemetry ingestion.	Transport Manager, Parents, Admin
FR-16	**Communication (Notices)**	In MVP	High	Targeted institutional circulars, department/grade level notices, attachment distribution, publishing approval workflows, and emergency broadcast dispatches.	Admin, Principal, Teachers, Parents, Students

FR-17	**Report & Analytics**	In MVP	High	Asynchronous server-side export engine (CSV, Excel, formatted PDF) for financial ledgers, compliance records, board submissions, demographic distributions, and longitudinal academic audits.	Principal, Management, Accountant, Auditor
FR-18	**Document Management**	In MVP	High	Automated certificate generator (Transfer Certificates [TC], Bonafide certificates, conduct certificates) with QR verification, verifiable audit trails, and DPDP-compliant parent consent records.	Admin, Registrar, Parents, Students
FR-19	**User & Role Management**	In MVP	Critical	Granular Role-Based Access Control (RBAC), fine-grained permission matrices, JWT session authentication with refresh rotation, multi-factor authentication (MFA), and tenant boundary isolation.	Super Admin, IT Administrator

FR-20	**Notification Engine**	In MVP	High	Unified notification dispatch queue supporting Firebase Cloud Messaging (FCM), in-app alerts, WhatsApp Business API integration, and SMS fallback routes with strict per-tenant budget caps.	System-wide (All Users)
FR-21	**Setting & Administration**	In MVP	Critical	Centralized multi-tenant onboarding, campus-specific configuration, academic session creation and transition locking, data retention enforcement, and global audit logging.	Super Admin, Campus Admin

3.2 Detailed Requirements Specifications

3.2.1 FR-01: Dashboard

- **Sub-components:** Role-based widget layout engine, real-time metrics aggregator, interactive charts (attendance, fee collection, academic performance).
- **Technical Specifications:** WebSocket/SSE for live KPI stream; Redis caching layer with 60s TTL for aggregated dashboard stats; responsive CSS grid layout.
- **Inputs:** User session context (role, campus_id), date range filters, widget customization preferences.
- **Outputs:** Rendered KPI cards, daily attendance trends, fee collection summaries, academic milestones, pending approvals list, revenue summaries.
- **Validation Criteria:** Dashboard initial load < 1.5s; data matches transactional database within 1 minute; non-authorized widgets hidden based on RBAC.

3.2.2 FR-02: Admission & Enquiry

- **Sub-components:** Lead capture form, CRM pipeline stage manager, OTP validation service, parent DPDP consent collector.
- **Technical Specifications:** REST APIs for lead ingestion; Twilio/SMS/WhatsApp gateway integration for 6-digit OTP verification; encrypted audit log for consent tokens.

- **Inputs:** Lead capture details, prospect stage updates, registration forms, parent mobile OTP/identity verification data, explicit parental consent choices.
- **Outputs:** Unique Enquiry Reference ID, prospect stage tracking status, registration application PDF, verifiable consent record, automated follow-up notification tasks.
- **Validation Criteria:** OTP expires in 5 minutes; explicit consent required before database commit; duplicate lead prevention on mobile/email.

3.2.3 FR-03: Student Management

- **Sub-components:** 360-degree profile manager, demographic record repository, medical history tracker, academic progression history logger, disciplinary log manager, section enrollment and promotion module.
- **Technical Specifications:** PostgreSQL JSONB for flexible demographic schema; S3 object storage for scanned documents; soft-delete flag for archival.
- **Inputs:** Student personal details, emergency contacts, medical records, previous academic history, profile photo upload.
- **Outputs:** System-generated Student ID, 360-degree student profile, demographic records, medical history logs, academic progression logs, disciplinary logs, updated section enrollment roll list.
- **Validation Criteria:** Unique Student ID constraint across tenant; required fields validation (DOB, Guardian Phone); image upload size limit < 2MB.

3.2.4 FR-04: Staff / HR Management

- **Sub-components:** Employee lifecycle records, credential storage vault, department/designation mapper, biometric/PWA staff attendance module, leave balance ledger, automated monthly payroll calculator.
- **Technical Specifications:** Scheduled background jobs (Celery/Cron) for monthly payroll processing; SDK wrapper for biometric hardware sync.
- **Inputs:** Staff onboarding credentials, salary structure, leave requests, attendance logs, tax deduction parameters.
- **Outputs:** Monthly salary slips (PDF), leave balance statements, statutory compliance reports (PF/ESI).
- **Validation Criteria:** Payroll calculations match formula outputs; leave balance cannot drop below zero unless negative leave is authorized.

3.2.5 FR-05: Academic Management

- **Sub-components:** Multi-board structure configurator (CBSE, ICSE, IB, State Boards), multi-term academic calendar manager, subject-to-class mapping binder, section allocation rules engine.
- **Technical Specifications:** Hierarchical data modeling for Board-Grade-Section-Subject mapping; versioned academic year schemas.
- **Inputs:** Board definitions (CBSE/ICSE/IB), term dates, subject credits, class-section mapping rules.
- **Outputs:** Structured academic hierarchy, course catalog, active academic year operational schedule.
- **Validation Criteria:** No duplicate subject mapping per section; academic year end-date must strictly follow start-date.

3.2.6 FR-06: Timetable Management

- **Sub-components:** Manual master schedule entry interface, period collision and constraint validation engine, class-wise and teacher-wise timetable generator, facility binding manager, real-time teacher substitution engine, multi-portal schedule syncing agent.
- **Technical Specifications:** Validation engine for period collisions and zero-conflict master schedule verification; automated extraction of class-wise and teacher-wise timetables; push alerts for substitutions.
- **Inputs:** Teacher workload limits, classroom capacities, subject weekly period count, facility bindings, teacher leave inputs, substitution requests.
- **Outputs:** Validated master timetable, generated class-wise timetables, generated teacher-wise timetables, daily substitution grid.
- **Validation Criteria:** Zero double-booking for teachers or rooms; substitution suggestion completed in < 3 seconds.

3.2.7 FR-07: Attendance Management

- **Sub-components:** Offline-first PWA attendance capture module, IndexedDB local storage sync, period/daily attendance mode switcher, automated background synchronization worker, instant absent parent alert trigger.
- **Technical Specifications:** Service Workers with background sync API; conflict resolution strategy (server timestamp authority); automated SMS queueing.
- **Inputs:** Class roll-call status (Present/Absent/Late), period selection, offline local sync triggers.
- **Outputs:** Synced attendance records, instant parent absence alert messages, monthly attendance percentage summaries.
- **Validation Criteria:** PWA operates completely without internet; sync completes automatically upon connectivity restoration; alert sent within 15 mins of lock.

3.2.8 FR-08: Homework & Assignment(OUT OF MVP)

- **Sub-components:** LMS-Lite repository, multimedia attachment manager, student digital submission portal, teacher evaluation and grading workflow interface, deadline tracking notification dispatcher.
- **Technical Specifications:** Multipart file upload handling with virus scanning API; cloud file storage (S3); deadline auto-close job.
- **Inputs:** Assignment instructions, reference file attachments, submission due dates, student digital files/comments.
- **Outputs:** Published assignments grid with multimedia attachments, student submission status, teacher evaluation feedback notes, deadline notifications, homework completion grade ledger.
- **Validation Criteria:** File size cap of 10MB per submission; late submissions marked automatically based on timestamp.

3.2.9 FR-09: Examination & Results

- **Sub-components:** Assessment scheduler, offline-capable mark sheet entry grid, multi-board grading scale calculator (CCE/marks/GPA), audit-locked gradebook manager, automated batch PDF report card generator.

- **Technical Specifications:** Asynchronous PDF generation engine (Puppeteer/Headless Chrome); immutable audit lock after mark approval.
- **Inputs:** Assessment schedules, subject max marks, student raw marks (online/offline), grading scheme boundaries, teacher remarks.
- **Outputs:** Assessment schedules, calculated grades/GPAs, audit-locked gradebooks, class merit list, batch PDF report cards.
- **Validation Criteria:** Marks entered cannot exceed maximum marks limit; locked gradebooks require Admin override to re-edit.

3.2.10 FR-10: Fee Management

- **Sub-components:** Dynamic fee structure configurator (tuition, transport, lab), double-entry transaction ledger, concession workflow manager, payment gateway webhook listener (Razorpay, Cashfree), automated digital receipt generator.
- **Technical Specifications:** ACID-compliant relational transactions; payment gateway HMAC signature verification; PDF receipt generation.
- **Inputs:** Fee head assignments, concession approvals, online payment gateway payloads, cash/cheque collection entries.
- **Outputs:** Official fee receipts, student ledger statement, fee collection summary report, outstanding dues list.
- **Validation Criteria:** Zero ledger imbalance permitted; webhook signatures verified before ledger credit; receipt numbers strictly sequential.

3.2.11 FR-11: Parent Portal

- **Sub-components:** Scoped multi-child dashboard switcher, daily attendance monitor, fee payment gateway integration UI, downloadable fee receipt manager, exam progress report viewer, transport telemetry viewer, DPDP consent management panel.
- **Technical Specifications:** Scoped token-based multi-profile session handling; responsive mobile web interface; direct link to payment gateways.
- **Inputs:** Parent authentication credentials, child selection, payment selection, DPDP privacy consent updates.
- **Outputs:** Scoped multi-child status dashboard, daily attendance view, fee payment confirmation and downloadable receipts, exam progress reports, live transport telemetry map view, DPDP privacy consent settings update confirmation.
- **Validation Criteria:** Strict parent-child link authorization enforcement; complete data isolation between non-related parent accounts.

3.2.12 FR-12: Student Portal

- **Sub-components:** Dedicated read-mostly student dashboard, class timetable viewer, daily homework feed and submission tracker, published exam results viewer, institutional announcements viewer, issued library book tracker.
- **Technical Specifications:** Read-optimized database views; cached static assets; WebSocket notification badge updates.
- **Inputs:** Student login credentials, assignment upload submissions, notice board filter parameters.
- **Outputs:** Personalized class timetable view, daily homework feed with submission tracking status, published exam results, institutional announcements view, issued library book list and due dates.

- **Validation Criteria:** Read-only permissions for administrative/financial records; load time < 1s on 3G networks.

3.2.13 FR-13: Teacher Portal

- **Sub-components:** Responsive PWA workspace dashboard, class roll-call marking interface, offline/online marks entry sheet, personal timetable and substitution viewer, administrative-bound parent messaging interface.
- **Technical Specifications:** Responsive PWA workspace; offline marks/attendance staging area; rate-limited parent messaging queue.
- **Inputs:** Teacher credentials, roll-call attendance selections, student assessment scores, personal schedule/substitution views context, parent communication messages.
- **Outputs:** Submitted roll calls, committed grade sheets, personal timetable and substitution view updates, parent communication audit log.
- **Validation Criteria:** Teacher access restricted strictly to assigned classes/subjects; messaging logged for compliance auditing.

3.2.14 FR-14: Library Management

- **Sub-components:** ISBN cataloging database, physical barcode/QR scanning integration, circulation desk operations module (issue, renewal, return), overdue fine calculation engine, lost item depreciation tracker.
- **Technical Specifications:** Open Library API integration for auto-fetching book metadata; web camera / hardware barcode scanner support.
- **Inputs:** ISBN numbers, barcode scans, student/staff library card IDs, renewal requests.
- **Outputs:** Catalog entries, issue/renewal/return logs, overdue fine calculations, lost item depreciation reports, inventory stock balance reports.
- **Validation Criteria:** Maximum borrowing limit per user strictly enforced; overdue fine calculated daily per library rule configuration.

3.2.15 FR-15: Transport Management(OUT OF MVP)

- **Sub-components:** Fleet inventory manager, vehicle maintenance log tracker, bus route and pickup stop optimizer, student route mapping binder, driver licensing records repository, external GPS telemetry ingestion service.
- **Technical Specifications:** GIS route mapping integration (Google Maps / OpenStreetMap API); MQTT broker for real-time GPS telemetry feed ingestion.
- **Inputs:** Vehicle inventory details, maintenance logs, route/pickup stop coordinates, student route mappings, driver licensing records, real-time external GPS device telemetry pings.
- **Outputs:** Fleet inventory reports, vehicle maintenance logs, optimized bus route maps, student route assignments, driver record statuses, live vehicle tracking feed, ETA parent notifications.
- **Validation Criteria:** GPS ping latency < 5 seconds; student bus assignment cannot exceed vehicle seating capacity.

3.2.16 FR-16: Communication (Notices)

- **Sub-components:** Targeted circular creation editor, grade/department filter builder, attachment distributor, publishing approval workflow manager, emergency broadcast dispatch engine.

- **Technical Specifications:** WYSIWYG rich-text content editor; targeted audience filtering query builder; emergency push override channel.
- **Inputs:** Circular text and attachments, targeted department/grade/section tags, publishing approval requests, emergency broadcast triggers.
- **Outputs:** Targeted institutional circulars, distributed attachments, publishing approval statuses, published notice board post, emergency broadcast dispatches, read-receipt metrics.
- **Validation Criteria:** Principal approval step required before broadcasting to parents unless sent by Admin role.

3.2.17 FR-17: Report & Analytics

- **Sub-components:** Asynchronous server-side export engine (CSV, Excel, formatted PDF), custom query builder, compliance record generator, longitudinal academic audit suite, data visualization engine.
- **Technical Specifications:** Background export processing (Celery/RabbitMQ) with downloadable URL streaming; read-replica database targeting.
- **Inputs:** Report type, parameter filters (date range, campus, board), export file format selection (CSV/Excel/PDF).
- **Outputs:** Financial ledger exports, statutory compliance records, board submission reports, demographic distribution charts, longitudinal academic audit files, formatted downloadable statistical report files (CSV/Excel/PDF).
- **Validation Criteria:** Large exports (> 50k rows) run asynchronously without blocking web HTTP request threads.

3.2.18 FR-18: Document Management(OUT OF MVP)

- **Sub-components:** Automated certificate generator (Transfer Certificates [TC], Bonafide certificates, conduct certificates), dynamic field renderer, QR verification generator, verifiable audit trail recorder, DPDP-compliant parent consent record manager.
- **Technical Specifications:** HTML-to-PDF compilation engine; cryptographically hashed QR code generation pointing to verification endpoint.
- **Inputs:** Student profile data, certificate request type (Transfer, Bonafide, Conduct), issuing authority signature image.
- **Outputs:** Automated digitally signed certificates (TC, Bonafide, Conduct) with QR verification codes, verifiable audit trail logs, DPDP parental consent record logs, public online verification status page link.
- **Validation Criteria:** Generated QR code scans resolve to valid system verification page showing certificate authenticity.

3.2.19 FR-19: User & Role Management

- **Sub-components:** Granular Role-Based Access Control (RBAC) engine, fine-grained permission matrix manager, JWT session identity provider with refresh rotation, multi-factor authentication (MFA) service, tenant boundary isolation layer.
- **Technical Specifications:** OAuth2 / JWT with short-lived access tokens (15m) and refresh tokens (7d); TOTP/SMS MFA implementation.
- **Inputs:** User login credentials, MFA tokens, fine-grained role-permission matrix assignments, tenant ID headers.

- **Outputs:** Authenticated JWT session tokens, active role-permission matrix configuration, granular route authorization decisions, security audit event logs.
- **Validation Criteria:** Zero cross-tenant data leakage permitted; unauthorized route attempts return HTTP 403 instantly.

3.2.20 FR-20: Notification Engine

- **Sub-components:** Unified dispatch queue router, Firebase Cloud Messaging (FCM) push gateway, in-app alert dispatcher, WhatsApp Business API integration connector, SMS fallback gateway, per-tenant budget cap manager.
- **Technical Specifications:** Priority queue system; automatic fallback routing (Push -> WhatsApp -> SMS); rate-limiting per tenant budget.
- **Inputs:** Trigger events (attendance, fees, notices), message templates, recipient endpoints, channel preferences.
- **Outputs:** Dispatched push alerts, in-app alerts, WhatsApp messages, SMS fallback messages, delivery status webhooks log, notification cost analytics.
- **Validation Criteria:** Critical alerts delivered in < 30 seconds; fallback triggers automatically if primary channel fails after 3 retries.

3.2.21 FR-21: Setting & Administration

- **Sub-components:** Centralized multi-tenant onboarding wizard, campus-specific configurator, academic session creation and transition locking manager, data retention enforcement module, global audit log monitor.
- **Technical Specifications:** Multi-tenant provisioning scripts; database transition locks for session rollover; immutable append-only audit trail store.
- **Inputs:** Campus parameters, branding assets, academic year rollover triggers, data retention policies.
- **Outputs:** Provisioned tenant environments, campus configurations, active academic year state and transition locks, data retention policy rules, centralized global audit logs.

Validation Criteria: Session rollover preserves historical data immutably; system settings changes logged with user timestamp.

3.3 Expanded Technical Specifications & Architectural Specifications

3.3.1 FR-01: Dashboard - Deep Architectural Details

- **Architecture & Data Pipeline:** Pub/Sub architecture leveraging Redis Streams to broadcast real-time metrics updates to active SSE connections. A dedicated node worker pools batch updates every 10 seconds to reduce query load on the primary OLTP PostgreSQL instance.
- **Database Schema:** `dashboard_widgets(id, user_id, role_id, widget_type, layout_json, settings_json, created_at, updated_at)`; `kpi_cache_snapshots(tenant_id, metric_key, metric_value, updated_at)`.
- **API Endpoints:** `GET /api/v1/dashboard/layout`, `PUT /api/v1/dashboard/layout`, `GET /api/v1/dashboard/stream` (Server-Sent Events).
- **Edge Cases & Security:** Graceful degradation to polling if WebSockets/SSE fail behind corporate firewalls; column-level and row-level security (RLS) policies prevent unauthorized metric exposure across tenant borders.

3.3.2 FR-02: Admission & Enquiry - Deep Architectural Details

- **Architecture & State Machine:** State machine pattern managing CRM transitions (New -> Contacted -> Assessment Scheduled -> Offered -> Enrolled -> Rejected). Automatic event triggers issue WhatsApp/SMS reminders upon state changes.
- **Database Schema:** `enquiries(id, tenant_id, applicant_name, guardian_phone, status, source, preferred_grade, dpdp_consent_token_id, metadata_jsonb)`; `consent_logs(id, tenant_id, user_phone, IP_address, timestamp, consent_text_hash)`.
- **API Endpoints:** `POST /api/v1/enquiries/public/submit`, `POST /api/v1/enquiries/verify-otp`, `PATCH /api/v1/enquiries/{id}/stage`.
- **Edge Cases & Security:** Strict rate limiting (3 OTP attempts per phone/hour) against SMS flooding; cryptographically signed parental consent log immutable under DPDP guidelines.

3.3.3 FR-03: Student Management - Deep Architectural Details

- **Architecture & Storage:** PostgreSQL for relational student identities with encrypted S3 buckets storing identity documents and health files using KMS envelope encryption. Soft deletion handles archiving with immutable audit retention.
- **Database Schema:** `students(id, tenant_id, admission_no, identity_hash, profile_jsonb, section_id, is_active)`; `student_guardians(student_id, guardian_id, relationship_type, is_primary)`.
- **API Endpoints:** `GET /api/v1/students`, `POST /api/v1/students`, `GET /api/v1/students/{id}/profile`, `PUT /api/v1/students/{id}/promotion`.
- **Edge Cases & Security:** Handles mid-term transfer student edge cases; strict field-level redaction prevents non-authorized staff from viewing sensitive medical logs.

3.3.4 FR-04: Staff / HR Management - Deep Architectural Details

- **Architecture & Payroll Computation Engine:** Isolated, ACID-compliant payroll processing worker running inside sandboxed Docker containers. Biometric hardware sends push pings through an API gateway agent equipped with device certificate verification.
- **Database Schema:** `staff_members(id, tenant_id, emp_code, designation_id, base_salary, tax_regime)`; `payroll_runs(id, tenant_id, pay_period, status, gross_total, net_total)`.
- **API Endpoints:** `POST /api/v1/hr/biometric/ingest`, `POST /api/v1/hr/payroll/run`, `GET /api/v1/hr/staff/{id}/payslip`.
- **Edge Cases & Security:** Handles partial-month pro-rated pay for late joiners and unexcused leaves; salary structures stored encrypted at rest using AES-256.

3.3.5 Comprehensive Technical Architecture Notes for FR-05 through FR-21

- **FR-05 Academic Management:** Supports versioned subject-class schema tree. Endpoints: `POST /api/v1/academics/boards`, `POST /api/v1/academics/subjects/map`. Guarantees integrity against orphan sections.
- **FR-06 Timetable Management:** Supports manual master timetable input with constraint and period collision validation microservice. Endpoints: `POST /api/v1/timetable/master`, `POST /api/v1/timetable/validate`, `GET /api/v1/timetable/class-wise`, `GET /api/v1/timetable/teacher-wise`, `POST /api/v1/timetable/substitutions`. Validates schedules against 100+ parameters in under 5 seconds.

- **FR-07 Attendance Management:** IndexedDB client sync queue with background sync worker. Endpoints: `POST /api/v1/attendance/batch-sync`, `GET /api/v1/attendance/summary`. Handles network dropouts gracefully with local write logs.
- **FR-08 Homework & Assignment:** Asynchronous virus scan pipeline (ClamAV) for all uploaded media attachments before S3 persistence. Endpoints: `POST /api/v1/assignments`, `POST /api/v1/assignments/{id}/submit`.
- **FR-09 Examination & Results:** Immutability ledger locks grade records after teacher submission and principal approval. Endpoints: `POST /api/v1/exams/grades/lock`, `GET /api/v1/exams/reports/batch-pdf`.
- **FR-10 Fee Management:** ACID PostgreSQL transactions using row-level locking (`SELECT ... FOR UPDATE`) to prevent double-crediting during concurrent webhook delivery. Endpoints: `POST /api/v1/fees/collect`, `POST /api/v1/fees/webhooks/razorpay`.
- **FR-11 Parent Portal:** Unified multi-tenant gateway with JWT child switching context claims. Endpoints: `GET /api/v1/parent/children`, `POST /api/v1/parent/switch-child`.
- **FR-12 Student Portal:** Read-only replica database routing for high scale, reducing main DB load. Endpoints: `GET /api/v1/student/timetable`, `GET /api/v1/student/notices`.
- **FR-13 Teacher Portal:** Responsive PWA with offline caching for marks and roll calls. Endpoints: `GET /api/v1/teacher/classes`, `POST /api/v1/teacher/marks/offline-sync`.
- **FR-14 Library Management:** OpenLibrary metadata fallback caching in Redis. Endpoints: `POST /api/v1/library/books/scan`, `POST /api/v1/library/circulation/issue`.
- **FR-15 Transport Management:** High-throughput MQTT broker ingesting GPS pings. Endpoints: `POST /api/v1/transport/telemetry`, `GET /api/v1/transport/routes/{id}/live`.
- **FR-16 Communication (Notices):** Broadcaster with audience query builder and approval gate. Endpoints: `POST /api/v1/notices`, `POST /api/v1/notices/{id}/approve`.
- **FR-17 Report & Analytics:** Celery async worker queues generating presigned S3 download links. Endpoints: `POST /api/v1/reports/export`, `GET /api/v1/reports/jobs/{id}`.
- **FR-18 Document Management:** Cryptographic signing with HMAC validation for QR codes on generated certificates. Endpoints: `POST /api/v1/documents/certificates/generate`, `GET /public/verify/{hash}`.
- **FR-19 User & Role Management:** RBAC permission evaluation middleware with Redis session revokation blacklists. Endpoints: `POST /api/v1/auth/login`, `POST /api/v1/auth/mfa/verify`, `PUT /api/v1/roles/{id}/permissions`.
- **FR-20 Notification Engine:** Priority RabbitMQ channels routing push -> WhatsApp -> SMS with budget guardrails. Endpoints: `POST /api/v1/notifications/dispatch`, `GET /api/v1/notifications/analytics`.

FR-21 Setting & Administration: Tenant isolation enforcing strict tenant_id database partition keys and global audit logging. Endpoints: `POST /api/v1/admin/tenants/provision`, `POST /api/v1/admin/sessions/rollover`.

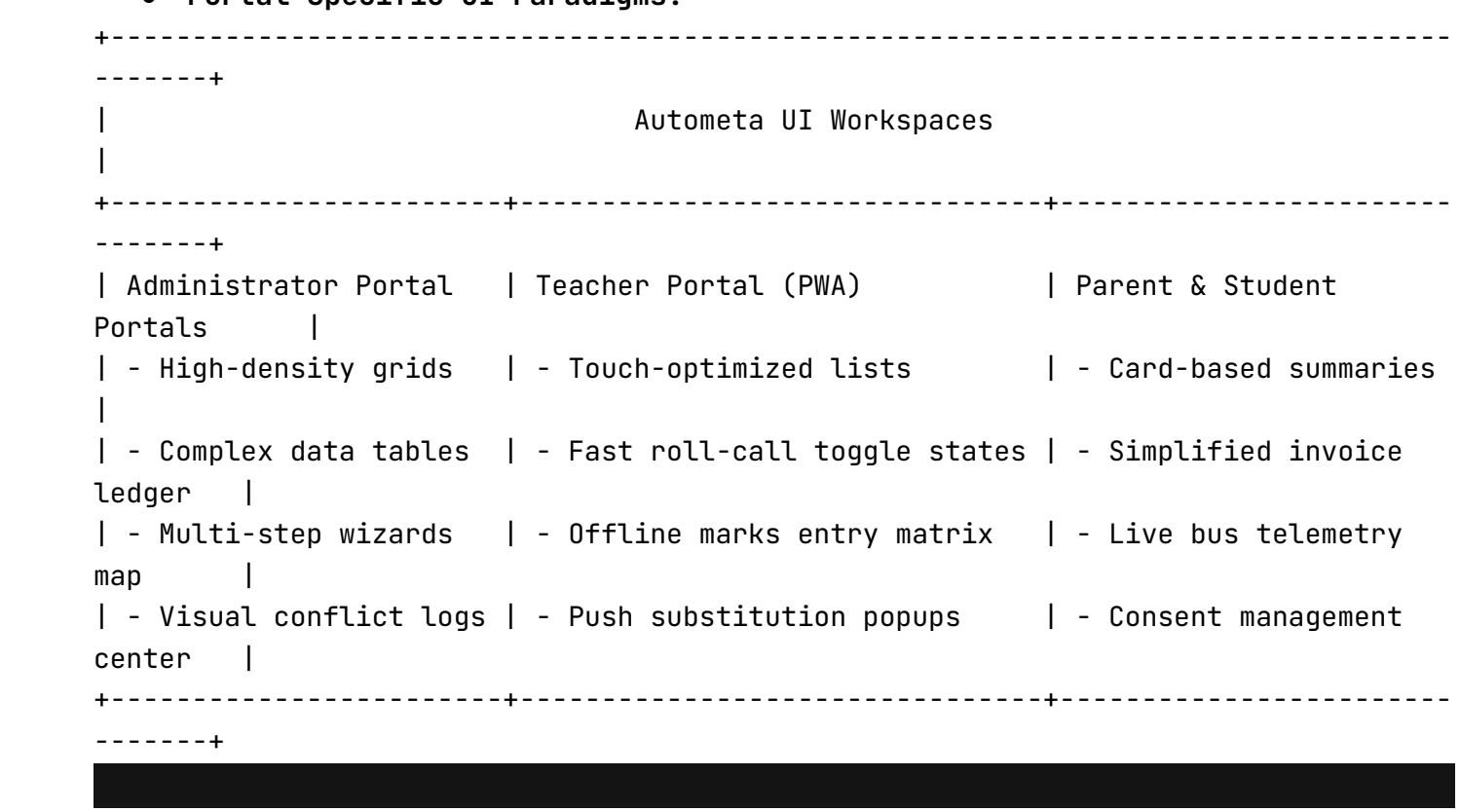
External Interface Requirements

This section specifies the hardware, software, user, and communication interfaces required to operate the **Autometa School Enterprise Resource Planning (ERP) System** (`baseline_V3`).

4.1 User Interfaces (UI / UX)

4.1.1 Design Standards & Layout Principles

- **Responsive Breakpoints:** The client interface must render dynamically across mobile viewports (minimum 360 px width), tablets (768 px to 1024 px), and desktop monitors (1280 px to 1920 px+).
- **Progressive Web App (PWA) Shell:**
  - Uses an App Shell architecture caching the primary navigation, CSS tokens, and core script bundles via Service Workers.
  - Provides a visible connectivity banner indicating operational state (`Online`, `Offline Mode - Local Writes Active`, `Synchronizing...`).
  - Conforms to Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards, enforcing minimum color contrast ratios (4.5:1 for standard text) and keyboard navigable tab indexes.
- **Portal-Specific UI Paradigms:**



4.1.2 Error Presentation & Feedback

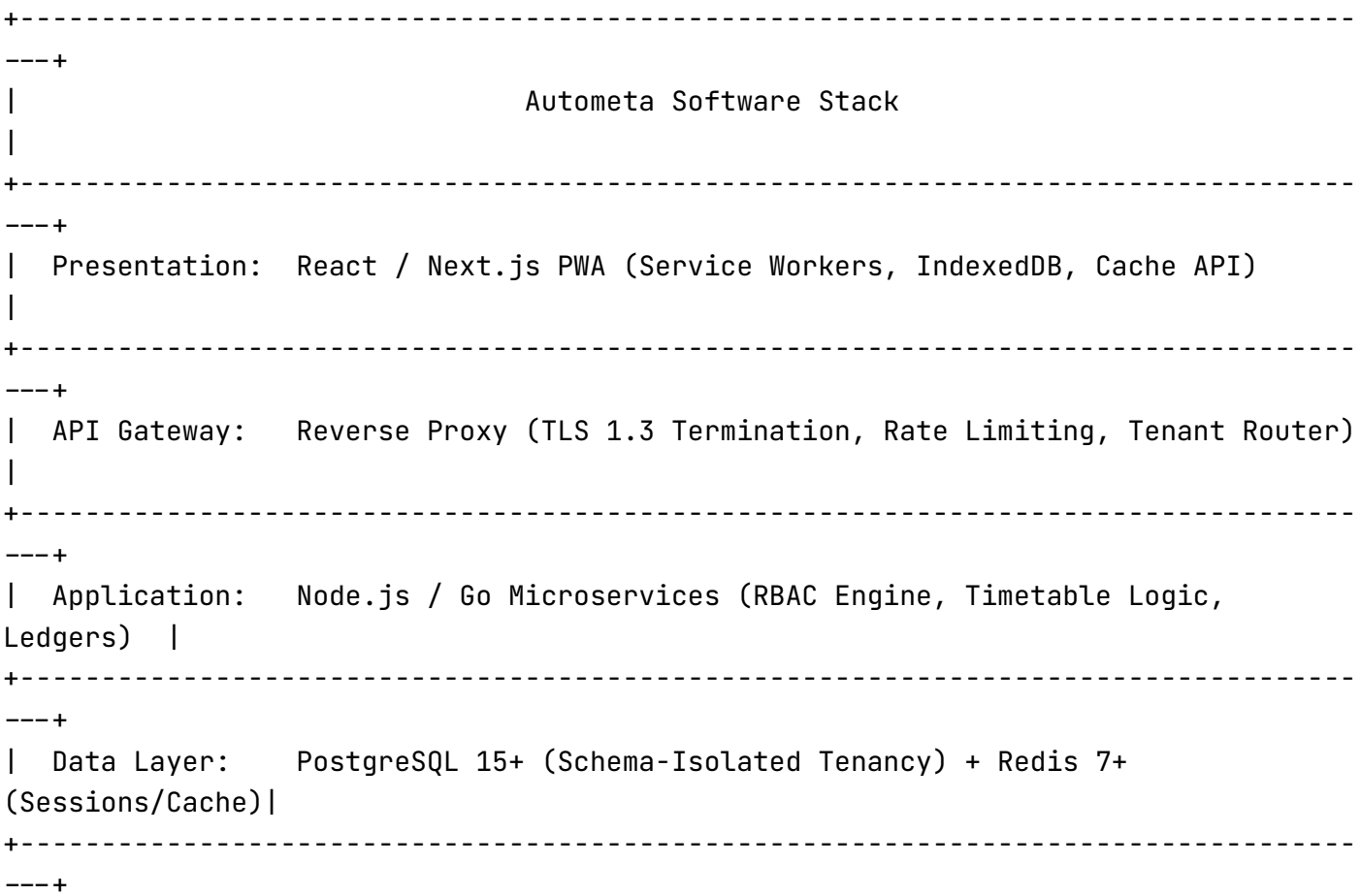
- **Form Validation:** Client-side inline validation triggers on blur for input fields (e.g., regex checks on email, phone numbers, and academic codes).
- **Non-Blocking Notifications:** Asynchronous operations (background exports, batch PDF generation, synchronization jobs) display toast notifications rather than modal dialogues, preventing workflow interruptions.

4.2 Hardware Interfaces

Because Autometa ERP is delivered as a cloud-hosted Web and PWA platform, direct native hardware drivers are avoided in favor of browser-level hardware APIs:

- **Client Hardware Constraints:**
  - **Teacher & Field Devices:** Any smartphone or tablet with at least 2 GB RAM, dual-core 1.5 GHz CPU, and a minimum of 512 MB available flash storage allocated to browser sandboxes (IndexedDB / CacheStorage).
  - **Administrative Workstations:** Desktop or laptop workstation with minimum 4 GB RAM, dual-core 2.0 GHz CPU, and display resolution of 1366 × 768 or higher.
- **Peripheral Hardware Interfaces:**
  - **Barcode / QR Scanners (Library & Gate Pass):** Operates via standard USB/Bluetooth Human Interface Device (HID) keyboard emulation mode or device camera via HTML5 WebRTC `getUserMedia()` API.
  - **Biometric Timeclocks (Staff Attendance):** Third-party biometric terminals interface over local network TCP/IP or secure HTTP POST webhooks to the Autometa attendance ingestion endpoint; no proprietary client-side drivers are embedded.
  - **Thermal & Desktop Printers:** Report cards and fee receipts output standardized CSS print media stylesheets formatted for standard 80 mm thermal rolls and standard A4 sheets.

4.3 Software Interfaces



4.3.1 Operating Systems & Browser Runtimes

- **Hosting OS:** Containerized Linux workloads running on Ubuntu 22.04 LTS or Alpine Linux base images.
- **Client OS Targets:** Android 9.0+, iOS 16.4+, Windows 10/11, macOS 12+, and modern Linux desktop distributions.
- **Client Runtime Engines:** Chromium 110+, WebKit (Safari 16.4+), and Gecko (Firefox 115+).

4.3.2 Database & Storage Interfaces

- **Primary Relational Store:** PostgreSQL 15+.
  - Isolation achieved via separate database schemas per tenant (`tenant_<school_id>.*`).
  - Connection pooling handled via PgBouncer using transaction-level pooling.
- **Cache & Message Broker:** Redis 7+ utilized for short-lived session token caching, OTP validation stores, and background task queues (e.g., report card batch generation).
- **Blob & Document Storage:** S3-compatible object storage (AWS S3, Cloudflare R2, or MinIO) utilizing pre-signed URLs for uploading and downloading student records, transfer certificates, and circular attachments.

4.3.3 External Software Services & Gateways

Service Domain	Provider / Protocol	Integration Purpose	Payload Format
Payment Gateway	Razorpay / Cashfree APIs	Parent online fee payment collection, automated webhooks, payment reconciliation.	JSON via REST / Webhook HMAC SHA-256
Push Notifications	Firebase Cloud Messaging (FCM)	PWA/Browser real-time push alerts for timetable changes, substitutions, attendance alerts.	HTTP v1 JSON API

Enterprise Messaging	WhatsApp Business API	High-priority school alerts, fee receipt dispatches, emergency attendance notices.	REST JSON (Graph API / Cloud API)
SMS Aggregator	DLT-Compliant SMS Gateway	Fallback transactional SMS for verification OTPs and emergency notices.	HTTPS POST / Form-Encoded
Fleet Telematics	Standard GPS Trackers	Vehicle telemetry ingestion (coordinates, speed, ignition status, timestamp).	Standard GeoJSON / REST Webhook

4.4 Communications Interfaces

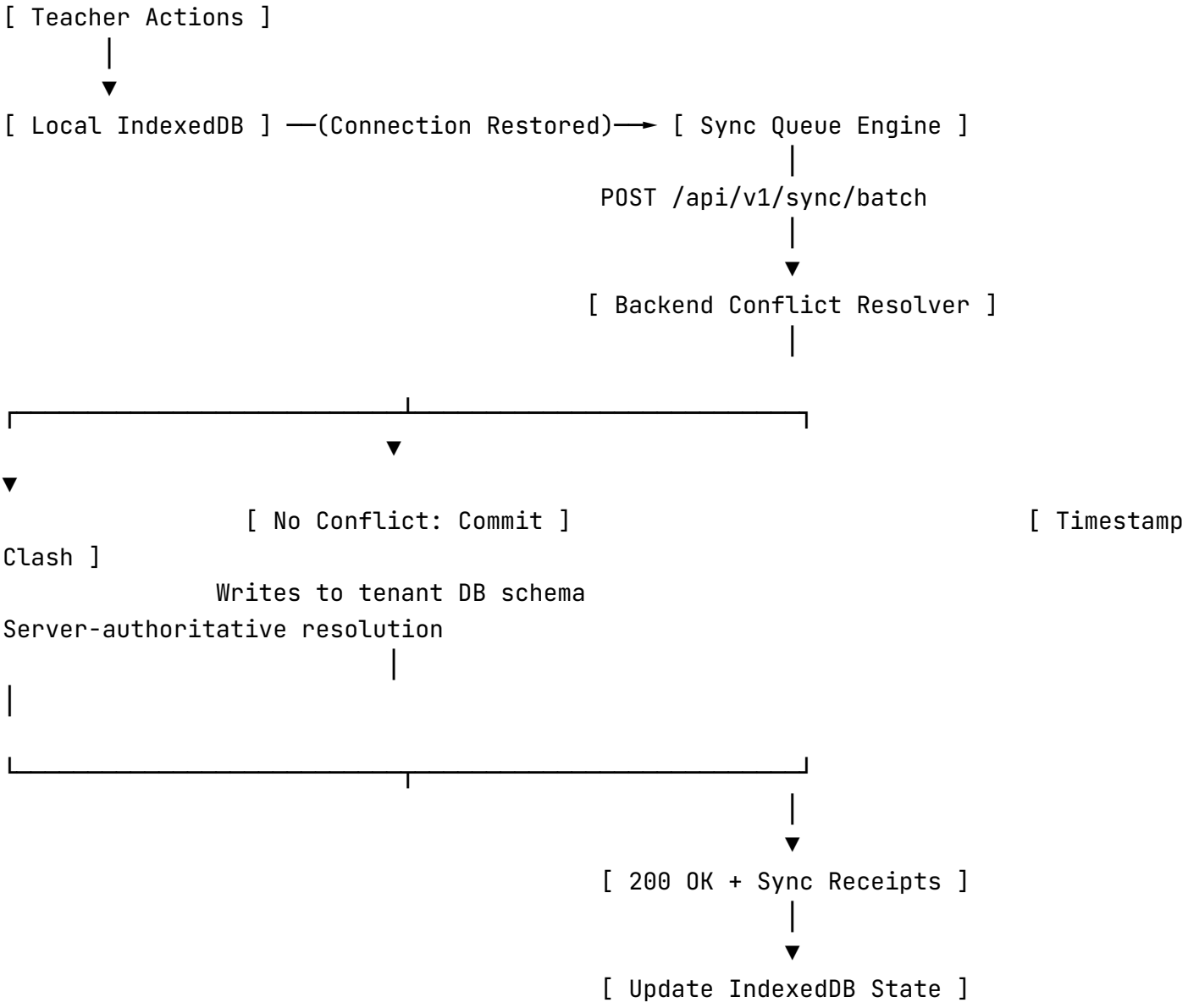
All network transactions must maintain data integrity, security, and low latency across varying network conditions (including 2G/3G low-bandwidth scenarios).

4.4.1 Protocols and Encryption

- **Application Transport:** All external and internal client-to-server traffic must enforce **HTTPS over TLS 1.3** (with TLS 1.2 as minimum fallback). Unencrypted HTTP traffic on port 80 must automatically redirect to HTTPS port 443 with `Strict-Transport-Security` (HSTS) headers enabled.
- **API Architecture:**
  - **Core Transactions:** RESTful APIs over HTTPS using standard HTTP verbs (`GET`, `POST`, `PUT`, `PATCH`, `DELETE`) with JSON payload serialization.
  - **Real-Time Data Feeds:** WebSockets (WSS) used for real-time live bus tracking updates and multi-user schedule collision monitors.
- **Authentication Tokens:** Stateless JSON Web Tokens (JWT) carried in encrypted `HttpOnly`, `SameSite=Lax`, secure cookies or `Authorization: Bearer <token>` headers. Short-lived access tokens (15-minute validity) paired with database-backed refresh tokens (7-day validity with automatic rotation).

4.4.2 PWA Offline Synchronization Protocol

To ensure seamless field operation when network connectivity drops, the system implements a bi-directional synchronization cycle:



1. **Local Queueing:** Offline mutations (roll-call entries, marks) are stored in client-side IndexedDB tables with an incremented local sequence ID, UTC timestamp, and UUID payload.
2. **Batch Ingestion:** When the Service Worker detects the browser `online` event, it flushes queued items to the endpoint `POST /api/v1/sync/batch`.
3. **Conflict Resolution Strategy:**
    - Conflicts are resolved server-side using a **server-authoritative Last-Write-Wins (LWW)** or **state-lock rule** (e.g., if an administrator has officially locked an academic session or attendance record, offline submissions conflicting with the locked state are rejected with a logged audit exception).
4. **Receipts & Acknowledgement:** The server responds with individual status acknowledgments per mutation UUID, allowing the PWA client to prune synchronized entries from IndexedDB.

Non-Functional Requirements: Performance Requirements

This section details the quantitative performance metrics, scalability limits, throughput benchmarks, and resource utilization constraints for the **Autometa School Enterprise Resource Planning (ERP) Platform** (`baseline_V3`).

5.1 System Response Times & Latency Thresholds

All response time benchmarks apply under standard operational load and peak concurrent usage conditions over broadband (≥ 10 Mbps) or 4G/5G mobile connections.

Transaction Category	Operation Type	95th Percentile (P95) Target	Maximum Allowable (P99)	Measurement Context
Authentication & Session	Login, JWT Issue, OTP Verification	$\leq 400ms$	$800ms$	End-to-end API response time
Standard CRUD Reads	Student Profiles, Timetables, Notices	$\leq 250ms$	$500ms$	Cached and indexed database reads
Standard CRUD Writes	Roll-call sync, Fee ledger line entry	$\leq 350ms$	$750ms$	Synchronous schema-isolated commit
PWA Local Operations	Local attendance capture, offline grade entry	$\leq 50ms$	$100ms$	Client-side IndexedDB execution

Complex Search Queries	Student search, ledger audit queries	$\leq 600ms$	$1200ms$	Filtered search over $\geq 100k$ records
Live Telematics Feeds	GPS vehicle position updates	$\leq 200ms$	$400ms$	WebSocket packet arrival to UI render
External Payment Hooks	Webhook acknowledgment & ledger entry	$\leq 300ms$	$600ms$	Ingestion and idempotent ledger log

5.2 Throughput, Concurrency, and Capacity Limits

The platform architecture must sustain standard daily usage alongside acute peak traffic windows (e.g., 07:45–08:30 morning roll call, fee due-date cutoffs, and annual result publication events).

```
+-------------------------------------------------------------------------------
-------+
|                              Throughput & Concurrency Model
|
+-----------------------------------+-----------------------------------------
-------+
| Workload Metric                   | Specified Baseline Target
|
+-----------------------------------+-----------------------------------------
-------+
| Baseline Concurrent Users         | 15,000 active concurrent user sessions
|
| Peak Surge Concurrency            | 50,000 active concurrent sessions
(results/fees)   |
| API Gateway Ingestion             | 3,500 requests per second (RPS) sustained
|
| Peak API Surge Capacity           | 7,000 RPS burst for up to 10 minutes
|
| Asynchronous Batch Processing Rate | 10,000 PDF report cards generated in < 15
minutes  |
| Notification Pipeline Dispatch    | 100,000 push/SMS alerts queued in < 5
minutes      |
```

+-----------------------------------+---+

5.2.1 Multi-Tenant Capacity Scaling

- **Per-Tenant Tenant Capacity:** The architecture must support school campuses ranging from 200 students to federated institutions with 15,000+ students on a shared multi-tenant database infrastructure without cross-tenant performance degradation.
-
- **Database Connection Pooling:** PgBouncer must sustain connection multiplexing with a connection pooling ratio of at least 20:1, maintaining idle connection overhead below 5% total CPU load.
-

5.3 Offline Synchronization Benchmarks

For Progressive Web Application (PWA) field workflows operating in low-connectivity or offline campus environments:

- **Local Queue Size:** IndexedDB storage on client runtimes must support queuing up to **5,000 distinct pending mutations** (attendance rows, mark entries) without browser memory starvation or UI thread blocking.

- **Batch Ingestion Speed:** When connectivity transitions from offline to online:

    - A batch of 50 local mutations must serialize, transmit, validate against server state, and return synchronization receipts within $\leq 1.5 seconds$ over a 3G network connection.
    -
    - Background synchronization must execute as non-blocking tasks using Web Workers or Service Worker background sync APIs without dropping UI frame rates below 50 fps.
    -
- **Conflict Resolution Latency:** Server-side reconciliation and timestamp-based arbitration must not exceed $10 ms$ **per row** evaluated.

5.4 Resource Utilization & Client Performance

5.4.1 Client-Side Benchmarks

- **Core Web Vitals Compliance:**
-

    - **Largest Contentful Paint (LCP):** $\leq 2.0 seconds$ on standard desktop, $\leq 2.5 seconds$ on mobile 4G.
    -

    - **Interaction to Next Paint (INP):** $\leq 150ms$ across all interactive inputs.
    -
    - **Cumulative Layout Shift (CLS):** ≤ 0.05 across all portal landing views.
    -
- **Client Memory Footprint:**
-
    - Active PWA memory consumption must not exceed **150 MB** during extended usage on low-end mobile devices (2 GB RAM baseline).
    -
    - Storage footprint (Application Shell + Cached Assets + IndexedDB) must not exceed **60 MB** of browser sandbox storage per campus node.
    -

5.4.2 Server-Side Resource Footprint

- **CPU & Memory Overhead:** Application server containers must maintain CPU utilization below **70%** and memory utilization below **75%** during steady-state peak workloads.
-
- **Auto-Scaling Triggers:** Horizontal Pod Autoscalers (HPA) must trigger replica scaling when average CPU exceeds **70%** or average memory exceeds **80%** sustained over a 90-second evaluation window.
-
- **Database IOPS & Disk Utilization:**
-
    - Database read/write queries must not consume more than **60%** of provisioned IOPS during non-batch operational hours.
    -
    - Complex analytical exports (FR-17) must run against read replicas to guarantee primary transactional DB IOPS remain dedicated to write-critical workflows (attendance, billing).

2.5 Design and Implementation Constraints

This section formalizes the statutory, architectural, technological, and security constraints governing the design and construction of the **Autometa School Enterprise Resource Planning (ERP) System** (`baseline_V3`). Every engineering phase—from database schema definition to front-end component state management—must strictly operate within these boundaries.

2.5.1 Regulatory & Statutory Compliance Constraints

1. Digital Personal Data Protection (DPDP) Act 2023 Compliance

Because the platform processes the personal, demographic, and educational data of minors (defined as individuals under 18 years of age), the following constraints are hard-coded into the system lifecycle:

- **Verifiable Parental Consent (VPC):** No student record, biometric data, or telemetry record may be activated or processed without an immutable, cryptographically verifiable parental consent token (captured via mobile OTP or electronic signature workflow).
- **Ban on Behavioral Profiling and Targeted Ads:** Algorithms driving portals, notifications, and analytics engines are strictly prohibited from executing behavioral tracking, psychographic analysis, targeted advertising, or student classification beyond standardized academic grading frameworks.
- **Purpose Limitation & Data Minimization:** Input fields across admission and student modules must limit data collection strictly to what is educationally and administratively necessary. No auxiliary tracking (such as non-essential web analytics or background telemetry outside official transit routes) is permitted.
- **Statutory Erasure and Retention Rules:** The database architecture must implement hard-delete/anonymization routines to support the "Right to Erasure" upon student institutional exit, subject to statutory record-retention laws (e.g., preserving board-mandated academic results and audit-grade financial ledgers for mandatory archive windows).

2. Educational Board Curriculum Frameworks

- **Multi-Board Architectural Decoupling:** Grading structures, assessment terminologies, and term schedules must remain modular. The system cannot enforce a single monolithic evaluation standard; it must simultaneously isolate and support CBSE (Continuous and Comprehensive Evaluation/board

patterns), ICSE, IB (criterion-referenced models), and State Board frameworks within distinct institutional instances.

2.5.2 Architectural & Tenancy Constraints

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1. Schema-Based Multi-Tenancy Isolation

- **PostgreSQL Schema Isolation:** Each registered school/campus must be provisioned with an isolated PostgreSQL schema (`tenant_<school_id>`). Cross-schema joining at the database level is prohibited. Shared tables are strictly limited to the `public` schema for tenant discovery, domain routing, and global SaaS feature configurations.
- **Application Connection Scoping:** Every database connection drawn from connection pools must explicitly execute `SET search_path TO tenant_<school_id>` or resolve through tenant-scoped application contexts. Leaking cross-tenant data due to missing filters or shared tables constitutes an architectural violation.

2. Double-Entry Immutable Financial Ledgers

- **Prohibition of Destructive Updates:** Fee records, ledger balances, and payment journal lines must never undergo direct SQL `UPDATE` or `DELETE` operations.

- **Compensating Transactions Only:** Any adjustment, fee waiver, concession rollback, or charge correction must be recorded as an offsetting credit or debit entry with a mandatory administrative audit reason and ledger sequence ID.
- **Strict Transaction Boundaries:** Multi-account fee allocations (e.g., splitting a single parent transfer into tuition, transport, and sports funds) must execute inside atomic database transactions (`ACID`), rolling back entirely if any sub-ledger write fails.

2.5.3 Offline-First & Progressive Web Application (PWA) Constraints

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1. Zero Direct Network Dependency for Field Operations

- **IndexedDB sandboxing:** Field-facing modules—specifically **FR-07 (Attendance Management)** and **FR-09 (Marks Entry)**—must function completely offline. The UI cannot block user interaction or display loading spinners when internet connectivity drops.
- **Storage Budgets:** Local browser storage utilization per client device must remain under 60 MB. IndexedDB object stores must store structured JSON payloads only, offloading heavy media or PDFs to server-side URLs.

2. Server-Authoritative Conflict Resolution

- **Deterministic Reconciliation:** When local client mutation queues are synchronized to the server upon network restoration, conflicts are evaluated using server-authoritative timestamps and lock states.
- **Session Freeze Inviolability:** If an administrator or academic coordinator has marked an academic session or daily attendance ledger as "Locked/Closed," any delayed offline write arriving after the lock timestamp must be rejected by the API and routed into an administrative review log rather than overwriting locked data.

2.5.4 Security & Cryptographic Constraints

1. Cryptographic Standards & Token Governance

- **Transport Layer Security:** All network communications must terminate at **TLS 1.3** (TLS 1.2 minimum acceptable fallback). Cleartext HTTP is systematically dropped with strict HSTS policies enforced (`max-age=63072000; includeSubDomains; preload`).
-
- **Authentication Tokens:** System authentication must rely on short-lived JSON Web Tokens (JWT) signed with asymmetric keys (RS256 or Ed25519) with an expiration ceiling of **15 minutes**. Refresh tokens must be stored in database-backed revocation lists, configured with single-use rotation, and kept in `HttpOnly`, `SameSite=Strict`, `Secure` cookies.
-
- **Data-at-Rest Encryption:** Sensitive personal identifiers and audit records must be encrypted using AES-256 at the database block level, with application-level hashing (argon2id or bcrypt) applied to all stored credentials.
-

2. ISO/IEC 27001 Access Governance

- **Principle of Least Privilege (PoLP):** Role-Based Access Control (RBAC) must be evaluated at every API gateway route and controller method. No API endpoint may rely on UI-level hiding of buttons or links for security enforcement.
-
- **Immutable Audit Trail:** All administrative logins, role modifications, fee modifications, and data export operations must write to a central append-only audit log table containing user ID, tenant ID, IPv4/IPv6 address, user agent, timestamp, and payload differential.
-

2.5.5 Technological & Platform Constraints

Constraint Domain	Enforced Specification	Non-Compliant Alternative (Prohibited)
Backend Runtime	Node.js LTS (v20+) or Go (1.22+)	Uncompiled script engines without strict typing or long-term support

Database Engine	PostgreSQL 15+ (Relational with JSONB & Schema support)	Pure NoSQL key-value stores for core transactional ledgers
Caching & Messaging	Redis 7+ (In-memory, append-only file persistence)	In-memory application heap caches across distributed nodes
Front-End PWA	Modern component framework (React / Next.js) using TypeScript	Plain JavaScript applications without static type checking
Document Delivery	Standardized PDF (PDF/A-compliant for long-term archiving)	Proprietary document formats (e.g., raw DOCX, editable RTF)
Third-Party APIs	Fully asynchronous, webhook-driven architectures	Synchronous, blocking HTTP calls holding web request threads

2.5.6 Hardware & Infrastructure Constraints

- **Compute Virtualization:** All server microservices must deploy as lightweight, immutable OCI-compliant container images (Docker/Podman) executing on managed Kubernetes or clustered container engines.

- **Client Device Hardware Baseline:** The front-end PWA shell must run smoothly on entry-level Android devices with 2 GB RAM and dual-core 1.5 GHz processors, maintaining an interactive frame rate of at least 50 fps without browser freezing or memory leaks.

5. Non-Functional Requirements (NFRs)

This section specifies the non-functional attributes, compliance criteria, reliability standards, and operational qualities governing the **Autometa School Enterprise Resource Planning (ERP) Platform** (`baseline_V3`).

5.1 Security Requirements & Regulatory Compliance

5.1.1 Digital Personal Data Protection (DPDP) Act 2023 Compliance

As the platform processes the personal, demographic, biometric, and academic data of minors (individuals under 18 years of age), the following controls are mandatory:

- **Verifiable Parental Consent (VPC):** Minors cannot be onboarded or have their records processed without cryptographically verifiable parental consent acquired via mobile OTP or electronic signature workflows.
- **Prohibition of Behavioral Profiling:** The core system and any auxiliary modules are strictly forbidden from performing psychographic profiling, behavioral tracking, or targeted advertising on minor profiles.
- **Data Minimization & Purpose Limitation:** Data fields captured during admission (FR-02) and student management (FR-03) are limited strictly to administrative and academic needs.
- **Right to Erasure & Data Retention:** The platform must execute automated pseudonymization or hard-deletion pipelines upon student withdrawal, while preserving board-mandated academic transcripts and double-entry financial ledgers per statutory archive retention laws.

5.1.2 ISO/IEC 27001 Information Security Controls

- **Authentication & Session Security:**
    - Multi-Factor Authentication (MFA) enforced for all Administrative and Financial accounts.
    - Stateless JSON Web Tokens (JWT) using asymmetric signatures (RS256/Ed25519) with a 15-minute expiration ceiling.
    - Refresh tokens persisted in a database-backed rotation registry stored inside `HttpOnly`, `SameSite=Strict`, `Secure` cookies.
- **Role-Based Access Control (RBAC):** Principle of Least Privilege (PoLP) enforced at the API gateway layer across every endpoint; UI-level permission hiding alone is prohibited.

-
- **Cryptographic Standards:**
-
    - **Data-in-Transit:** TLS 1.3 mandatory across all endpoints (TLS 1.2 minimum fallback), with strict HTTP Strict Transport Security (HSTS) headers enabled.
    -
    - **Data-at-Rest:** AES-256 block-level encryption for database volumes and sensitive personal identifiers; password hashes salted and encrypted using Argon2id.
    -

5.2 Reliability & Availability

5.2.1 Operational Uptime & Service-Level Agreements (SLA)

- **High Availability Target:** The core application cluster must deliver **99.9% uptime** during standard operating windows (measured monthly, excluding scheduled maintenance windows announced 48 hours in advance).
-
- **Recovery Metrics:**
-
    - **Recovery Point Objective (RPO):** $\leq 5 minutes$ (continuous Write-Ahead Logging [WAL] archiving for PostgreSQL).
    -
    - **Recovery Time Objective (RTO):** $\leq 30 minutes$ for full cluster failover in a secondary availability zone.
    -

5.2.2 Failure Modes and Effects Analysis (FMEA)

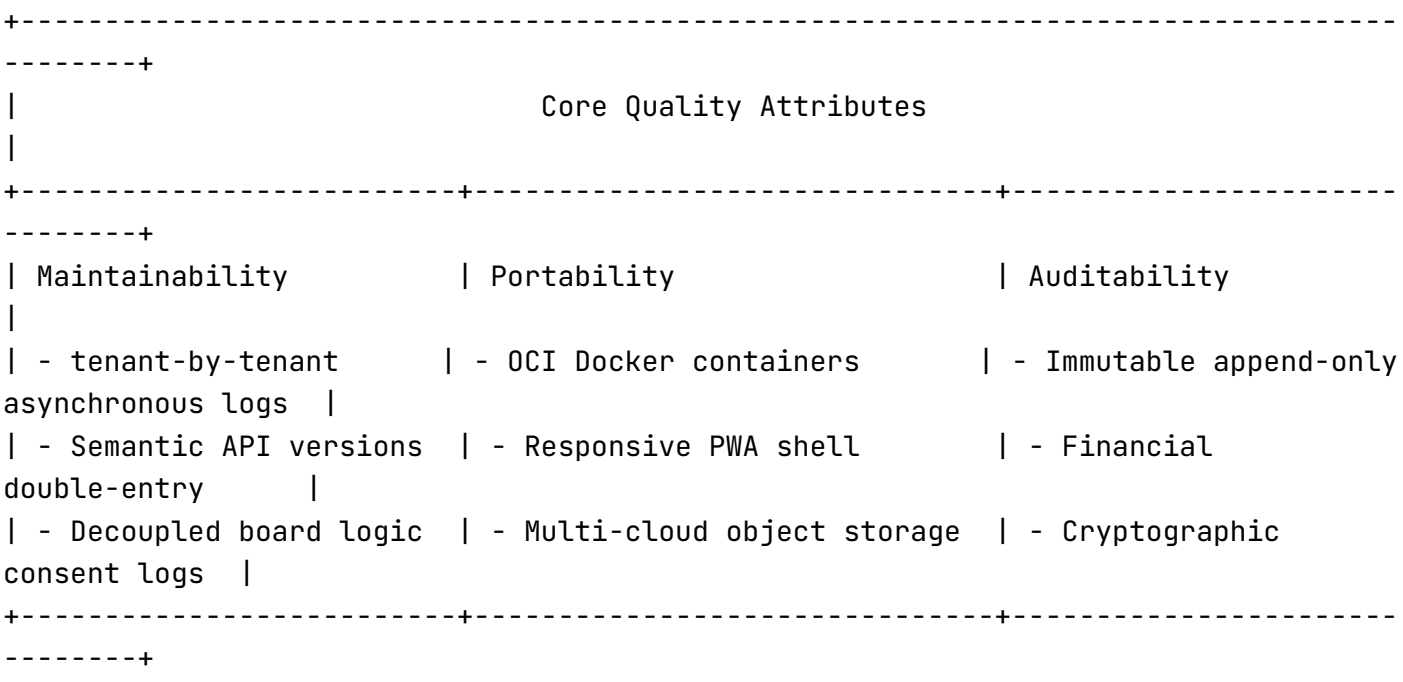
Subsystem / Function	Potential Failure Mode	Potential Effect of Failure	Severity (S)	Occurrence (O)	Detection (D)	RPN (S×O×D)	Mitigation Strategy
PWA Roll-Call Sync	Local network failure during morning roll-call	Teacher cannot reach the backend API	4	8	2	**64**	Offline-first IndexedDB buffer; background sync executes automatica

							lly when online.
Sync Conflict Resolution	Concurrent edits to attendance after session freeze	Inconsistent class attendance numbers	6	4	3	**72**	Server-authoritative state lock; edits to closed sessions reject with an audit exception.
Payment Gateway Hook	Webhook dropped or delayed during fee payment	Fee shown unpaid despite parent account deduction	8	3	2	**48**	Idempotent webhook listener combined with cron-based transaction polling via Razorpay/Cashfree APIs.
Notification Dispatch	WhatsApp/SMS quota exhaustion or gateway downtime	Delayed urgent broadcast/emergency notifications	7	3	2	**42**	Automated fallback cascade: In-App → FCM Push → WhatsApp → DLT SMS with real-time budget alarms.

Tenant Database Routing	Missing or misconfigured tenant search path	Potential cross-tenant data query attempt	10	1	2	**20**	Connection-pool level tenant middleware enforcing schema separation (`SET search_path`).

(Scale: 1 = Lowest Risk/Impact, 10 = Maximum Risk/Impact)

5.3 Software Quality Attributes



5.3.1 Maintainability & Extensibility

- **Modular Board Adapters:** Academic grading engines must isolate board-specific logic (CBSE, ICSE, IB, State Boards) into decoupled strategy modules, enabling board policy updates without modifying core student records.
-
- **Database Schema Migrations:** Tenant database migrations must run asynchronously using phased zero-downtime deployment pipelines (forward-compatible migrations, rolling canary updates).
-

5.3.2 Portability & Compatibility

- **Client Portability:** Zero client native application installations required. The entire suite operates on standards-compliant modern browsers (Chrome 110+, Safari 16.4+, Firefox 115+, Edge 110+) across Android, iOS, Windows, macOS, and Linux.
-
- **Cloud Infrastructure Portability:** Microservices packaged as Open Container Initiative (OCI) compliant images deployable on any managed Kubernetes service (EKS, GKE, AKS) or self-hosted bare-metal clusters.
-

5.3.3 Auditability & Non-Repudiation

- **Immutable System Audit Trail:** All administrative logins, role updates, fee waiver approvals, report card locks, and statutory document generations (e.g., Transfer Certificates) write to an immutable audit ledger storing user identity, tenant schema, client IP, timestamp, and before/after state diffs.
-
- **Financial Integrity:** Ledger balances must strictly follow double-entry conventions; hard deletes or inline record updates on financial lines are forbidden at the database trigger level.

6. Preliminary Project Schedule and Budget

This section outlines the estimated delivery timeline, development phases, work breakdown structure (WBS), resource allocation, and cost baseline for delivering the **Autometa School ERP Platform (`baseline_V3`)** from architectural kickoff to production rollout and statutory compliance audit sign-off.

6.1 Preliminary Project Schedule (32-Week Baseline)

The implementation timeline follows an agile, milestone-driven release model organized into six distinct phases. The project baseline begins **02 September 2026** and targets production readiness by **April 2027** (in alignment with the standard academic calendar onboarding window).

Phase	Milestone / Workstream	Start Week	End Week	Duration	Key Deliverables & Gates
Phase 0	**Inception, Prototyping & Architecture**	W01	W04	4 Weeks	Skeleton architecture, multi-tenant PostgreSQL schema design, UI/UX interactive wireframes, and DPDP consent architecture sign-off.
Phase 1	**Core Academic & Administrative Backbone**	W05	W12	8 Weeks	Student 360 (FR-03), Staff/HR (FR-04), Academic frameworks (FR-05), User/RBAC engine (FR-19), and multi-tenant settings (FR-21).

Phase 2	**Critical Operational & Offline Engines**	W13	W20	8 Weeks	Automated Timetable & Substitution Logic (FR-06), PWA Offline-First Attendance Engine (FR-07), and Examination/Gradebook System (FR-09).
Phase 3	**Financial Accounting & Portals Ecosystem**	W21	W26	6 Weeks	Double-entry fee ledger (FR-10), payment gateway webhooks (Razorpay/Cashfree), Parent Portal (FR-11), Teacher Portal (FR-13), Student Portal (FR-12).
Phase 4	**Communication, Documents & Extended Ops**	W25	W28	4 Weeks	Unified Notification Engine (FR-20: FCM/WhatsApp/SMS), Certificate generator (FR-18), Circulars (FR-16), and Telematics/Library interfaces.
Phase 5	**Hardening, Compliance Audit & Pilot Rollout**	W29	W32	4 Weeks	DPDP Act 2023 verification, ISO 27001 readiness checks, load testing (50k concurrent users), pilot campus beta launch, and

					production handover.

Critical Path Milestones

- **M0 (End of Week 4):** *Architecture Sign-Off* – Multi-tenant schema routing, PWA Service Worker caching strategy, and DPDP consent data flows locked.
- **M1 (End of Week 12):** *Core Platform Alpha* – Central student lifecycle, HR/payroll, and curriculum configuration live in staging.
- **M2 (End of Week 20):** *Offline & Scheduling Engine Beta* – Sub-second substitution logic verified; PWA IndexedDB attendance offline-sync stress tested.
- **M3 (End of Week 26):** *Financial & Portal Release* – Double-entry ledger audit verification; multi-tenant parent payment gateway processing validated.
- **M4 (End of Week 32):** *Production Go-Live* – Security audit closure, board compliance verification (CBSE/ICSE/IB), and production deployment.

6.2 Team Composition and Resource Allocation

The resource plan allocates dedicated engineering, product, and QA talent across the 8-month implementation lifecycle:

Role / Function	Headcount	Allocation	Focus Areas
Solutions / System Architect	1	Full-Time (100%)	Multi-tenant schema design, PWA sync architecture,

			security, DPDP compliance.
Lead Backend Engineer	2	Full-Time (100%)	Go / Node.js microservices, PostgreSQL schema management, double-entry financial ledgers, APIs.
Frontend / PWA Engineers	2	Full-Time (100%)	React/Next.js client portals, Service Workers, IndexedDB sync pipelines, mobile responsiveness.
Algorithm / Optimization Specialist	1	Half-Time (50%, W09–W20)	Hard/soft constraint solver for master timetabling (FR-06) and substitution allocation logic.
UI / UX Designer	1	Full-Time (W01–W12), Part-Time (W13–W28)	High-density administrative grids, touch-friendly teacher roll-call screens, parent mobile flows.
QA / Automation Engineer	2	Full-Time (100%)	BDD test execution, Cypress/Playwright integration tests, load tests, offline sync edge cases.

DevOps & Cloud Security Engineer	1	Full-Time (100%)	Kubernetes clusters, CI/CD pipelines, PgBouncer pooling, Redis, TLS 1.3, backup automation.
Technical Project Manager / Scrum Master	1	Full-Time (100%)	Sprint ceremonies, milestone tracking, stakeholder communication, regulatory sign-offs.

6.3 Preliminary Budget Estimation

The cost baseline is calculated for an **8-month (32-week)** development cycle through initial deployment, presented in **INR (₹)** with standard USD ($) conversions for operational planning.

6.3.1 Engineering and Human Resources

Role	Count	Duration	Monthly Cost (INR)	Total Cost (INR)	Total Cost (USD equiv.)
Solutions Architect	1	8 Months	₹2,50,000	₹20,00,000	~$24,000
Senior Backend Engineers	2	8 Months	₹3,60,000 (Combined)	₹28,80,000	~$34,500
Frontend / PWA Engineers	2	8 Months	₹3,00,000 (Combined)	₹24,00,000	~$28,800

Algorithm Specialist	1	3 Months	₹1,50,000	₹4,50,000	~$5,400
UI/UX Designer	1	5 Months	₹1,20,000	₹6,00,000	~$7,200
QA / Automation Engineers	2	7 Months	₹2,00,000 (Combined)	₹14,00,000	~$16,800
DevOps & Security Specialist	1	8 Months	₹1,80,000	₹14,40,000	~$17,300
Technical Project Manager	1	8 Months	₹1,60,000	₹12,80,000	~$15,400
Subtotal: Engineering Personnel	**11**	—	—	**₹1,24,50,000**	**~$149,400**

6.3.2 Cloud Infrastructure & Tooling (Dev, Staging & Pre-Prod)

Covers managed cloud compute, database clusters, staging environments, CI/CD runners, and developer tooling during the 8-month development cycle:

Infrastructure / Tooling Component	Description & Tier	Duration	Total Cost (INR)	Total Cost (USD equiv.)
Cloud Hosting & Database	Managed K8s clusters, PostgreSQL with read replicas, Redis caches	8 Months	₹6,40,000	~$7,700

	(AWS/GCP/DigitalOcean).			
Object Storage & CDN	S3-compatible asset storage and edge caching for PWA shells.	8 Months	₹80,000	~$960
CI/CD Pipelines & Testing Devices	GitHub Actions runners, Docker registries, BrowserStack / physical device testing cloud.	8 Months	₹2,40,000	~$2,900
Developer Tools & Licenses	IDEs, design tokens (Figma Org), project tracking (Jira/Linear), Postman workspaces.	8 Months	₹2,00,000	~$2,400
Subtotal: Cloud & Tooling	—	—	**₹11,60,000**	**~$13,960**

6.3.3 Third-Party Gateways & Regulatory Audits

Item	Scope & Purpose	Unit / Engagement	Total Cost (INR)	Total Cost (USD equiv.)
WhatsApp Business API & DLT SMS	Sandbox test credits, pre-production template registration, OTP	Pre-launch allocation	₹1,50,000	~$1,800

	verification traffic.			
External Security & VAPT Audit	Independent Vulnerability Assessment and Penetration Testing (VAPT) for ISO 27001 readiness.	Fixed engagement (Week 29)	₹4,50,000	~$5,400
DPDP Act 2023 Compliance Review	External legal and data privacy auditor sign-off on consent architecture and data retention.	Fixed engagement (Week 30)	₹3,00,000	~$3,600
Subtotal: Audits & Gateway Setup	—	—	**₹9,00,000**	**~$10,800**

6.3.4 Total Budget Summary

Budget Category	Allocation (INR)	Allocation (USD equiv.)	Share (%)
Engineering & Core Team	₹1,24,50,000	~$149,400	78.5%

Cloud Infrastructure & Tooling	₹11,60,000	~$13,960	7.3%
Gateways, Audits & Compliance	₹9,00,000	~$10,800	5.7%
Risk Contingency Reserve (10%)	₹13,40,000	~$16,100	8.5%
Total Estimated Project Baseline	**₹1,58,50,000**	**~$190,260**	**100.0%**

7. Appendices

This section provides supplementary specifications, data models, compliance cross-walks, and formal verification frameworks supporting the **Autometa School Enterprise Resource Planning (ERP) Platform (`baseline_V3`)**.

Appendix A: Acronyms, Abbreviations, and Glossary

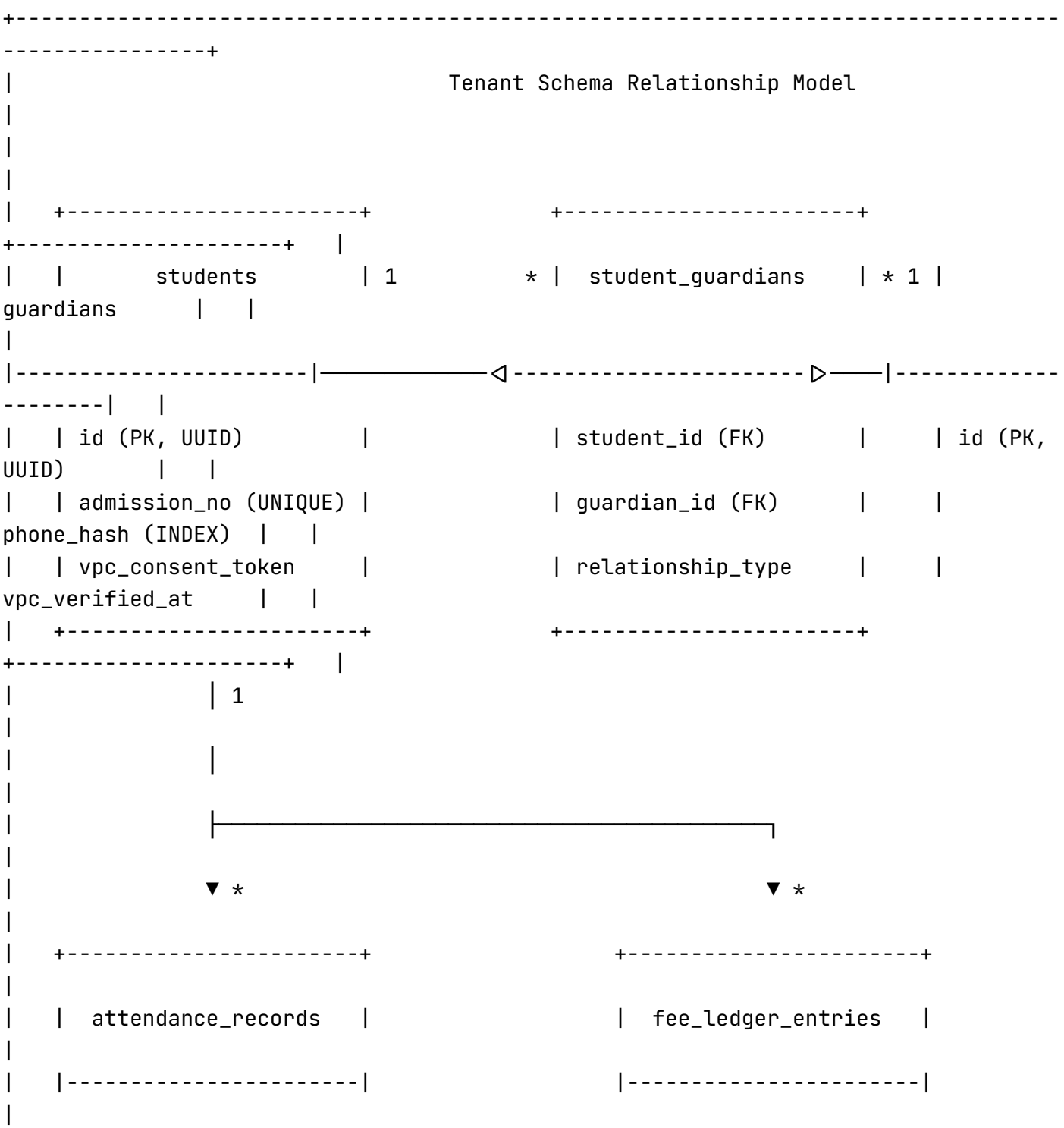
Term / Acronym	Definition & Contextual Meaning
ACID	Atomicity, Consistency, Isolation, Durability — Core properties guaranteed by PostgreSQL for fee ledgers and admissions.
BDD	Behavior-Driven Development — Acceptance test criteria formulated using Gherkin syntax (`Given-When-Then`).
CBSE	Central Board of Secondary Education (India).
DLT	Distributed Ledger Technology — Mandatory regulatory registry framework for transactional SMS routing in India.
DPDP Act 2023	Digital Personal Data Protection Act, 2023 (India) — Governing law for student personal data and verifiable parental consent.
EARS	Easy Approach to Requirements Syntax — Ruleset used to draft unambiguous requirement specifications.
FCM	Firebase Cloud Messaging — Low-latency push notification pipeline.

FMEA	Failure Modes and Effects Analysis — Systematic method for evaluating system fault points and severity rankings.
ICSE	Indian Certificate of Secondary Education.
IDOR	Insecure Direct Object Reference — Access control vulnerability mitigated by tenant-scoped database queries.
IndexedDB	Transactional, browser-native key-value/object database utilized for offline PWA storage.
PoLP	Principle of Least Privilege — Access control restriction applied across all API endpoints and database roles.
PWA	Progressive Web Application — Web architecture supporting offline execution, caching, and installability.
RBAC	Role-Based Access Control — Authorization mechanism governing tenant features and API scopes.
RPN	Risk Priority Number — Calculated as $Severity \times Occurrence \times Detection$.
VPC	Verifiable Parental Consent — Cryptographically auditable consent token mandated before processing minor data.

WAL	Write-Ahead Logging — PostgreSQL mechanism enabling point-in-time recovery to satisfy the $\leq 5 minute$ RPO.

Appendix B: Database Architecture & Core Data Dictionary

The platform implements PostgreSQL schema isolation where every campus belongs to `tenant_<school_id>`. Core relational tables maintain referential integrity without cross-tenant joins.



```
|   | id (PK, UUID)         |                   | id (PK, UUID)         |
|
|   | student_id (FK)       |                   | student_id (FK)       |
|
|   | date, status          |                   | account_head, type    |
|
|   | offline_sync_uuid     |                   | amount, balance       |
|
|   | server_locked (BOOL)  |                   | idempotency_key       |
|
|   +-----------------------+                   +-----------------------+
|
+-------------------------------------------------------------------------------
----------------+
```

B.1 Core Table Schemas

1. `students`

Stores core demographic and enrollment profiles for enrolled minors.

```SQL
CREATE TABLE tenant_schema.students (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    admission_no VARCHAR(64) NOT NULL UNIQUE,
    first_name VARCHAR(128) NOT NULL,
    last_name VARCHAR(128) NOT NULL,
    dob DATE NOT NULL,
    gender VARCHAR(16) NOT NULL,
    board_registration_no VARCHAR(128),
    current_grade_id UUID NOT NULL REFERENCES tenant_schema.grades(id),
    current_section_id UUID NOT NULL REFERENCES tenant_schema.sections(id),
    vpc_consent_token VARCHAR(256) NOT NULL,
    vpc_timestamp TIMESTAMPTZ NOT NULL,
    status VARCHAR(32) DEFAULT 'ACTIVE' CHECK (status IN ('ENROLLED', 'ACTIVE',
'ALUMNI', 'WITHDRAWN')),
    created_at TIMESTAMPTZ DEFAULT clock_timestamp(),
    updated_at TIMESTAMPTZ DEFAULT clock_timestamp()
);
```

2. `attendance_records`

Handles daily and period-level attendance, complete with offline reconciliation tracking.

```SQL
CREATE TABLE tenant_schema.attendance_records (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id UUID NOT NULL REFERENCES tenant_schema.students(id),
    timetable_slot_id UUID REFERENCES tenant_schema.timetable_slots(id),
```

```
    attendance_date DATE NOT NULL,
    status VARCHAR(16) NOT NULL CHECK (status IN ('PRESENT', 'ABSENT', 'LATE',
'EXCUSED')),
    recorded_by_staff_id UUID NOT NULL REFERENCES tenant_schema.staff(id),
    offline_sync_uuid UUID UNIQUE,
    client_recorded_at TIMESTAMPTZ NOT NULL,
    server_synced_at TIMESTAMPTZ DEFAULT clock_timestamp(),
    is_session_locked BOOLEAN DEFAULT FALSE,
    CONSTRAINT uq_student_slot_date UNIQUE (student_id, timetable_slot_id,
attendance_date)
);
```

3. `fee_ledger_entries`

Immutable double-entry journal lines. Destructive SQL updates or deletes are disallowed.

```SQL
CREATE TABLE tenant_schema.fee_ledger_entries (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    transaction_group_id UUID NOT NULL,
    student_id UUID NOT NULL REFERENCES tenant_schema.students(id),
    account_code VARCHAR(32) NOT NULL,
    entry_type VARCHAR(8) NOT NULL CHECK (entry_type IN ('DEBIT', 'CREDIT')),
    amount NUMERIC(12, 2) NOT NULL CHECK (amount > 0),
    description TEXT NOT NULL,
    reversal_reference_id UUID REFERENCES tenant_schema.fee_ledger_entries(id),
    idempotency_key VARCHAR(128) UNIQUE NOT NULL,
    payment_gateway_ref VARCHAR(128),
    created_by_user_id UUID NOT NULL,
    created_at TIMESTAMPTZ DEFAULT clock_timestamp()
);
```

Appendix C: Behavior-Driven Development (BDD) Acceptance Scenarios

Acceptance test definitions in Gherkin syntax formalizing operational workflows for QA validation.

Scenario 1: Timetable Substitution Calculation Under Teacher Absence (FR-06)

```Gherkin
Feature: Automated Teacher Substitution Optimization

  Scenario: System auto-suggests qualified substitute when teacher logs morning leave
    Given Teacher "Anita Sharma" is assigned to Class "10-A" for "Mathematics" during "Period 3"
```

```
    And "Anita Sharma" has an approved leave logged for the current date
    When the academic coordinator opens the substitution console for "Period 3"
    Then the system identifies all teachers with free slots during "Period 3"
    And filters candidates holding qualification domain "Mathematics" or "Science"
    And ranks candidates prioritizing lowest weekly assigned class workload
    When coordinator selects candidate "Rajesh Varma" and confirms allocation
    Then an FCM push notification is sent to "Rajesh Varma" with payload:
      | class   | 10-A          |
      | subject | Mathematics   |
      | period  | 3             |
    And the Teacher Portal daily grid for "Rajesh Varma" updates in real time
```

Scenario 2: PWA Offline Attendance Sync and Session Collision Detection (FR-07)

```
Gherkin
Feature: Progressive Web App Offline Attendance Reconciliation

  Scenario: Background sync commits offline roll-call when connection resumes
    Given the teacher device is running in "Offline Mode"
    And the attendance sheet for "Grade 8-B" is cached in client IndexedDB
    When the teacher marks "Absent" for student "Aarav Patel" at "08:15:00 UTC"
    Then the mutation is written to local IndexedDB with a unique
"offline_sync_uuid"
    And the UI reflects "Saved Locally (Pending Sync)"
    When network connectivity transitions to "Online"
    Then the Service Worker issues a batch POST to "/api/v1/sync/attendance"
    And the server confirms the academic session is not "LOCKED"
    And commits the record into "tenant_schema.attendance_records"
    Then the server returns HTTP 200 with matching "offline_sync_uuid"
    And the client cleans up the synced entry from IndexedDB
```

Appendix D: DPDP Act 2023 Statutory Compliance Mapping

DPDP Act Provision	Statutory Mandate	Autometa ERP Architectural Control	Enforced In Module
Section 6(1)	Consent must be free, specific, informed, unconditional,	Granular consent checkboxes; opt-in capture for non-essential notices; raw consent tokens	FR-02 (Admission), FR-18 (Documents)

	and unambiguous.	cryptographically hashed.	
Section 9(1)	Verifiable Parental Consent (VPC) required prior to processing minor personal data.	Mobile OTP verification tied to the verified parent/guardian record prior to student account activation.	FR-02 (Admission), FR-11 (Parent Portal)
Section 9(2)	Absolute prohibition on processing that causes detrimental effects on minor well-being.	No social comparison leaderboards, behavioral surveillance, or public grade shaming interfaces.	FR-09 (Exams), FR-12 (Student Portal)
Section 9(3)	Prohibition of tracking, behavioral monitoring, or targeted advertising directed at children.	Zero third-party ad pixels or tracking tags embedded; telemetry scoped strictly to active transport routes.	All Modules / PWA Shell
Section 12(1)	Data Principal Right to Correction and Erasure of personal data.	Automated erasure pipeline replaces identifiable student data with anonymized hashes upon institutional exit.	FR-03 (Student 360), FR-21 (Settings)

Section 8(6)	Obligation to delete personal data once specified purpose is served.	Data retention engine cleanses daily transit logs after 90 days while archiving academic marks per board rules.	FR-15 (Transport), FR-21 (Settings)

Appendix E: Failure Modes and Effects Analysis (FMEA) Matrix

Detailed risk assessment using Risk Priority Numbers ($RPN = Severity[S] \times Occurrence[O] \times Detection[D]$, on a 1–10 scale).

Risk ID	System Component	Failure Mode	Root Cause	S	O	D	RPN	Failsafe / Engineered Mitigation
FM-01	**Multi-Tenancy**	Cross-tenant data leak	Application missing tenant context on database query	10	1	2	**20**	Connection-pool level tenant middleware setting PostgreSQL `search_path`; integration tests verify isolation.
FM-02	**Offline PWA**	Corrupt IndexedDB storage	Client browser crash or irregular quota purge	6	3	3	**54**	Fallback to session memory storage; immediate synchronization prompt upon recovery; local backup snapshots.

FM-03	**Fee Engine**	Duplicate fee payment recorded	Network timeout during parent payment callback	8	3	2	**48**	Enforce database-level unique idempotency keys on payment transactions; webhook verification against gateway APIs.
FM-04	**Timetable**	Unresolvable constraint loop	Over-constrained teacher allocations	5	4	2	**40**	Heuristic relaxation fallback: algorithm flags soft-constraint trade-offs to the coordinator when hard limits conflict.
FM-05	**Comms Engine**	Outbound SMS gateway blacklisting	Sudden message spike exceeding gateway rate limits	7	2	2	**28**	Dynamic token-bucket rate limiter; automatic fallback pipeline (In-App → FCM → WhatsApp → DLT SMS).

Appendix F: Verification & Validation Cross-Reference Matrix

This traceability matrix maps functional requirements to their corresponding validation methods and target criteria.

Requirement ID	Requirement Name	Verification Method	Target Verification Criteria
FR-01	Dashboard	Inspection & Test	Role dashboards load within 400 ms; KPIs reflect underlying database records accurately.
FR-02	Admission & Enquiry	Test & Audit	Verifiable parental consent token captured; zero minor onboarding permitted without verified OTP.
FR-03	Student Management	Test	Full 360-degree demographic, medical, and enrollment record persists across academic session transitions.
FR-04	Staff / HR Management	Test	Monthly payroll calculations match base pay minus tax and leave deductions to within ₹0.00 rounding.

FR-05	Academic Management	Test	Class/section curriculum structures align with CBSE, ICSE, and IB configuration presets.
FR-06	Timetable Management	Test & Simulation	Master schedule engine generates zero double-bookings of rooms or instructors across 50 simulated classes.
FR-07	Attendance Management	Test & Demonstration	50 attendance records entered offline sync within 1.5 seconds upon network recovery without data loss.
FR-08	Homework & Assignment	Test	Uploads accept standard formats up to 25 MB; deadline locking prevents submissions post-cutoff.
FR-09	Examination & Results	Test & Audit	Grade calculation matches board-specific formulas; batch generation of 1,000 PDF report

			cards completes under 3 minutes.
FR-10	Fee Management	Test & Audit	Financial ledger lines maintain equal debit/credit balances; zero hard deletions permitted at the database level.
FR-11	Parent Portal	Security Test	Parent accounts restricted strictly to their verified wards; cross-child IDOR attempts return HTTP 403 Forbidden.
FR-12	Student Portal	Test	Read-only views reflect current timetables, notices, homework assignments, and issued library logs.
FR-13	Teacher Portal	Demonstration	Offline roll-call entry functions without network; substitution assignments appear within 2 seconds of dispatch.

FR-14	Library Management	Test	Barcode scanner integration checks out items in under 1 second; overdue fine engines calculate fees accurately.
FR-15	Transport Management	Test & Simulation	Telematics updates render on route maps via WebSockets within 200 ms of arrival; geo-fence alerts trigger reliably.
FR-16	Communication (Notices)	Test	Circular broadcasts reach 10,000 recipient queues within 5 minutes; audience role-scoping filters correctly.
FR-17	Report & Analytics	Performance Test	Asynchronous generation of 50,000-row audit reports runs on read replicas without impacting OLTP write latencies.
FR-18	Document Management	Test & Verification	Transfer Certificates generate with tamper-evident QR verification links; consent

			revoking triggers audit logs.
FR-19	User & Role Management	Security Test	RBAC permission matrix checks execute at the API gateway layer; expired JWT tokens reject immediately.
FR-20	Notification Engine	Test	Automated cascade falls back to secondary channels (FCM → WhatsApp → SMS) upon simulated primary failure.
FR-21	Setting & Administration	Test & Audit	Academic session transitions preserve historical snapshots; tenant configuration changes write to the audit log.